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Question 1

Not answered

Marked out of 1.00

How many episodes of mania and/or depression should be seen in 12 months time to diagnose rapid cycling bipolar disorder?

Select one:

- 3
- 4
- 2
- 6
- 8

Explanation:

'Rapid cycling' in bipolar disorder is defined as four or more episodes per year.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 328.

The correct answer is: 4

Question 2

Not answered

Marked out of 1.00

The Global Assessment of Functioning Scale is a rating scale measuring overall psychosocial functioning. What is the maximum score in the GAF scale?

Select one:

- 7
- 100
- 10
- 60
- 80

Explanation:

The Global Assessment of Functioning Scale (GAF) is a rating scale measuring overall psychosocial functioning. The maximum score on the GAF is 100.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 96.

The correct answer is: 100

Question 3

Not answered

Marked out of 1.00

The term simple schizophrenia was added as to the categories of existing schizophrenias by

Select one:

- Bleuler
- Kraepelin
- Hecker
- Angst
- Liddle

Explanation:

In 1896, Emil Kraepelin described and separated the two major forms of insanity on the basis of different symptoms, course, and outcome. The first, 'manic-depressive insanity', had a relapsing and remitting course, with full recovery after each episode. The second grouped together catatonia, hebephrenia, and the paranoid psychoses under the term 'dementia praecox', which had a progressive, deteriorating course. Eugen Bleuler added 'simple schizophrenia' to Kraepelin's subtypes and did not consider hallucinations, delusions, and catatonic symptoms to be necessary for the diagnosis. He believed the disorder to be due to a 'loosening of associations' between psychic functions, with fundamental symptoms being thought disorder, blunting/incongruity of affect, autism, and ambivalence.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 182.

The correct answer is: Bleuler

Question 4

Not answered

Marked out of 1.00

Which of the following is correct with regard to a diagnosis of post schizophrenic depression?

Select one:

- The patient must not have a previous history of depression
- The patient must not be on any antipsychotic medication
- Some positive symptoms must be present along with depression
- It can be diagnosed only 12 months after the last psychotic episode
- It is classified along with psychotic disorders

Post-schizophrenic depression was a category in the prior version of ICD (ICD-10 F20.4) where it was described as a type of schizophrenia along with other psychotic disorders. While the general criteria for schizophrenia must have been met within the past twelve months, they must not be met in full at present. But one of either positive/negative symptoms/formal thought disorder/catatonia symptoms must be present. The depressive symptoms must be sufficiently prolonged, severe and extensive to meet criteria for at least a mild depressive episode. The patient may or may not be on treatment with antipsychotics or antidepressants. Previous history of depression does not preclude a diagnosis. But since ICD-11 has introduced a symptom specifier to replace the subtypes of schizophrenia. These specifiers include positive, negative, depressive, manic, psychomotor and cognitive deficits. A patient has to be constantly assessed for the presence or absence of these specifiers, their longitudinal course, response to treatment and prognosis. The specifier's clinical severity can vary between patients and in the same patient at different times.

Ref: Valle R. Schizophrenia in ICD-11: Comparison of ICD-10 and DSM-5. Revista de Psiquiatría y Salud Mental (English Edition). 2020 Apr 1;13(2):95-104.

The correct answer is: It is classified along with psychotic disorders

Question 5

Not answered

Marked out of 1.00

Which of the following features seen in a person suffering from grief reaction 1 year after the loss suggests depression?

Select one:

- Mummification
- Guilt regarding omission of care
- Physical health problems
- Preoccupation with the relative
- Anniversary reaction

Signs of difficulty in assimilating the loss vary from the slight (e.g., expecting the person to return home from work and setting two places for dinner) to the extreme of mummification (e.g., retaining a child's room exactly as it was on the day of death for years). These signs may be harbingers of depression if they persist for more than one year. Mummification was first described by Gorer in 1965.

The correct answer is: Mummification

Question 6

Not answered

Marked out of 1.00

In order to make a ICD-11 diagnosis of delusional disorder, the delusions must be present for at least

Select one:

- 3 days
- 1 month
- 6 months
- 2 weeks
- 3 months

Explanation:

Delusional disorder is characterised by the development of a delusion or set of related delusions, typically persisting for at least 3 months and often much longer, in the absence of a depressive, manic, or mixed mood episode. The delusions are variable in content across individuals, but typically stable within individuals, although they may evolve over time. Other characteristic symptoms of schizophrenia (i.e. clear and persistent hallucinations, negative symptoms, disorganised thinking, or experiences of influence, passivity, or control) are not present, although various forms of perceptual disturbances (e.g. hallucinations, illusions, misidentification of persons) thematically related to the delusion are still consistent with the diagnosis. Apart from actions and attitudes directly related to the delusion or delusional system, affect, speech, and behaviour are typically unaffected.

Ref: International Statistical Classification of Diseases and Related Health Problems (11th ed. ICD-11; World Health Organization, 2020)

The correct answer is: 3 months

Question 7

Not answered

Marked out of 1.00

The most common co-morbid psychiatric condition associated with panic disorder is

Select one:

- Alcohol use disorder
- Schizophrenia
- Agoraphobia
- Borderline personality disorder
- Social phobia

Explanation:

Common comorbidities with panic disorder include agoraphobia (community surveys: 30-50%; psychiatric clinics: 75%), depressive disorder (anywhere from 40-80% of all patients, as estimated by various studies), other anxiety and related disorders (up to 50%, e.g. social phobia, specific phobia), alcohol (up to 30%) and other substance use disorders, and bipolar disorder (20%). Some medical conditions also share comorbidity with panic disorder (COPD, irritable bowel syndrome, migraine, mitral valve prolapse, hypertension, cardiomyopathy, stroke).

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 369.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 406.

The correct answer is: Agoraphobia

Question 8

Not answered

Marked out of 1.00

A person drinks alcohol as soon as he wakes up in the morning. During working hours, he again drinks in the lunch time and after getting home he needs to have at least 6 units of alcohol. His behaviour is best described as

Select one:

- Compulsive behaviour
- Alcohol dependence
- Rumination
- Impulsivity
- Obsessions

Explanation:

Alcohol dependence is characterised by harmful use of alcohol and an established dependence syndrome. Usually this includes a daily stereotyped drinking pattern, with increased tolerance, withdrawal features on abstinence, and 'relief drinking' (i.e. further drinking to alleviate the effects of withdrawals).

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 600.

The correct answer is: Alcohol dependence

Question 9

Not answered

Marked out of 1.00

Which of the following enzymes is the most sensitive marker of thiamine deficiency?

Select one:

- Alcohol dehydrogenase
- Lactate dehydrogenase
- Transketolase
- Aldehyde dehydrogenase
- Alpha ketoglutarate

Explanation:

Thiamine is an essential nutrient required by all tissues for the assembly and efficient functioning of different enzymes involved in the utilisation of carbohydrates for cellular function. Both the neurons and glial cells in the brain are affected by thiamine deficiency and have been implicated in the development of Wernicke encephalopathy as well as other forms of alcohol-related brain injury. Thiamine is metabolised by the liver to its biologically active form thiamine pyrophosphate (TPP). TPP is a vital co-enzyme involved in carbohydrate metabolism, and deficiency of thiamine can lead to decreased functioning of enzymes involved in the tricarboxylic acid cycle, pentose phosphate pathway (transketolase), and branched-chain amino acid metabolism. Transketolase is the most sensitive marker of thiamine deficiency; its activity can be reduced by up to 90% in areas of the brain that have a high sensitivity to thiamine levels. A decline in transketolase is correlated with the development of ataxia.

ef: Chandrakumar A et al. Review of thiamine deficiency disorders: Wernicke encephalopathy and Korsakoff psychosis. Journal of basic and clinical physiology and pharmacology. 2019 Mar 1;30(2):153-62.

The correct answer is: Transketolase

Question 10

Not answered

Marked out of 1.00

Length of time cannabis can be detected in the urine of a person who has used it only once is

Select one:

- 3 months
- 24 hours
- 1 month
- 7 days
- 3 days

Explanation:

Single use of cannabis can be detected in the urine for 3 days. With chronic, heavy use, it can be detected for up to one month.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 631.

The correct answer is: 3 days

Question 11

Not answered

Marked out of 1.00

Which of the following reflexes may NOT be exaggerated in upper motor neuron lesions?

Select one:

- Abdominal reflex
- Ankle jerk reflex
- Biceps reflex
- Knee jerk reflex
- Jaw jerk reflex

Deep tendon reflexes are exaggerated. Abdominal reflex is not a deep tendon reflex; it is a superficial reflex.

The correct answer is: Abdominal reflex

Question 12

Not answered

Marked out of 1.00

Which of the following syndromes is named after the person who initially described them?

Select one:

- Othello syndrome
- Fregoli syndrome
- Munchaussen syndrome
- Rabbit syndrome
- Cotard syndrome

Explanation:

Cotard syndrome is a presentation of psychotic depressive illness seen particularly in elderly people. There is a combination of severely depressed mood with nihilistic delusions and/or hypochondriacal delusions. The patient may state that they are already dead and should be buried, that their insides have stopped working and are rotting away, or that they have stopped existing altogether. It is named after the French neurologist Jules Cotard, who first described it in 1880.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 105.

Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Cotard syndrome

Question 13

Not answered

Marked out of 1.00

Which of the following is the most important discriminating feature between mild cognitive impairment and dementia?

Select one:

- Educational attainment
- Duration of symptoms of memory disturbances
- Family history of Alzheimer
- Activities of daily living
- MMSE score

Explanation:

Mild cognitive impairment (MCI) represents the preclinical, transitional stage between healthy ageing and dementia. MCI is separated from mild dementia by an absence of global intellectual deterioration and the preservation of activities of daily living (ADL). Recent clinical guidelines on MCI put forth by the National Institute on Aging-Alzheimer Association (NIA-AA) are as follows:

1.Concern regarding a change in cognition, from the patient, an informant, or a skilled clinician

2.Impairment in one or more cognitive domain(s) relative to a person's age and educational attainment

3.Independence in functional abilities/activities of daily living (although they may take more time or be performed less efficiently)

4.Not demented

Ref: Anderson ND. State of the science on mild cognitive impairment (MCI). CNS spectrums. 2019 Feb;24(1):78-87.

The correct answer is: Activities of daily living

Question 14

Not answered

Marked out of 1.00

You are interviewing a newly admitted patient in your ward. When asked what he had for breakfast, he says 'cereals'. Then when you asked 'what is your name?' he says 'cereals'. When you asked 'what is his date of birth?' he continues to say 'cereals'. The most likely symptom exhibited here is

Select one:

- Mannerism
- Automatism
- Perseveration
- Stuttering
- Stereotypy

Explanation:

Perseveration is characterised by continuing with a verbal response or action that was initially appropriate after it ceases to be apposite. It is associated with organic brain disease and is occasionally seen in schizophrenia.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 115.

The correct answer is: Perseveration

Question 15

Not answered

Marked out of 1.00

Which one of the following is best suited for a detailed assessment of global cognitive functioning?

Select one:

- Mini-COG
- Clock drawing test
- MMSE
- Montreal cognitive assessment guide
- Addenbrooke's cognitive examination

Explanation:

When time permits, or the clinical presentation is more complex, the Addenbrooke's Cognitive Examination (3rd edition) (ACE-III) provides a more detailed, 100-item, clinician-administered bedside test of cognitive function. Questions cover five areas of function: attention and orientation, memory, verbal fluency, language, and visuospatial awareness. The ACE-III has a reported sensitivity of 94% and a specificity of 89% for dementia, with a cut-off score of 88/100.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 86.

The correct answer is: Addenbrooke's cognitive examination

Question 16

Not answered

Marked out of 1.00

Which of the following is not a feature of Kubler-Ross model of grief?

Select one:

- Agitation
- Bargaining
- Denial
- Depression
- Acceptance

Explanation:

Elisabeth Kubler-Ross, a psychiatrist and thanatologist, made a comprehensive and useful organisation of reactions to impending death. She proposed the following five stages (which can be remembered using the acronym DABDA):

Stage 1: Shock and Denial

Stage 2: Anger

Stage 3: Bargaining

Stage 4: Depression

Stage 5: Acceptance

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 847.

The correct answer is: Agitation

Question 17

Not answered

Marked out of 1.00

Inability to verbalize one's strong emotions is termed as

Select one:

- Anhedonia
- Avolition
- Apathy
- Ambivalence
- Alexithymia

Explanation:

Alexithymia is the inability to describe one’s subjective emotional experiences verbally. It may be a personality characteristic but is also associated with somatisation.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 102.

The correct answer is: Alexithymia

Question 18

Not answered

Marked out of 1.00

Which of the following may indicate poor response to psychotherapeutic interventions?

Select one:

- alexithymia
- insomnia
- apraxia
- anhedonia
- amnesia

Explanation:

The literature shows that alexithymia is a potent predictor of resistance to treatment. Available data tend largely to correlate low baseline, and/or post-treatment of levels of alexithymia, and/or improvement in levels of alexithymia over the course of treatment, with a more favourable outcome of therapeutic interventions in several mental disorders. However, the presence of numerous discordant findings prevents any firm conclusions as to the actual impact produced by alexithymia on the treatment of psychiatric disorders both in general and in specific diagnostic groups.

Ref: Pinna F et al. The impact of alexithymia on treatment response in psychiatric disorders: a systematic review. Frontiers in psychiatry. 2020 Apr 16;11:311.

The correct answer is: alexithymia

Question 19

Not answered

Marked out of 1.00

A patient with schizophrenia protrudes his tongue repeatedly in and out when asked to do it once. He is exhibiting which of the following symptoms?

Select one:

- Chorea
- Stereotypy
- Ambivalence
- Hemiballism
- Ambitendency

Explanation:

Ambitendency can be regarded as a mild variety of negativism. Some consider this as an alternating mixture of automatic obedience and negativism. The presence of opposing tendencies for motor acts may also be regarded as a 'motor' form of ambivalence. The patient appears to be in conflict with opposing tendencies to action resulting in a 'jack in the box' feature of catatonia.

In ambitendency, the patient makes a series of tentative movements that do not reach the intended goal when they are expected to carry out a voluntary action. For example, when the examiner puts his hand out to shake hands, the patient moves their hand towards the examiner's hand, stops, starts moving the hand, stops, and so on, until the hand finally comes to rest without touching the examiner's hand.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 102.

Casey P, Kelly B. Fish's clinical pathology: Signs and symptoms in psychiatry, 4th Edition. 2019. p. 95.

The correct answer is: Ambitendency

Question 20

Not answered

Marked out of 1.00

Ben is worried that he has to shout 'Willy Wonka' whenever he sees an attractive woman, despite his best efforts to keep calm. This is best termed as

Select one:

- A first rank symptom of schizophrenia
- A mental compulsion
- An obsessional impulse
- Somatic passivity
- A made act

Explanation:

The essential feature of an obsession is that it appears against the patient's will. It naturally follows that we can only call a mental event an obsession if it is normally under the control of the patient and can be resisted by the patient. Thus, we have obsessional mental images, ideas, fears, and impulses, but not obsessional hallucinations or moods. Obsessional impulses may be impulses to touch, count or arrange objects, or impulses to commit antisocial acts. It is generally unusual for the patient to carry out an obsessive impulse.

Ref: Casey P, Kelly B. Fish's clinical pathology: Signs and symptoms in psychiatry, 4th Edition. 2019. p. 46.

The correct answer is: An obsessional impulse

Question 21

Not answered

Marked out of 1.00

Mr Y has a cardiac arrest. He is in the resuscitation room and experiences seeing himself outside of his body. This experience is called

Select one:

- Somatic hallucination
- Lilliputian hallucination
- Reflex hallucination
- Autoscopic hallucination
- Extracampine hallucination

Explanation:

Autoscopy (also known as 'phantom mirror image') is the experience of seeing a visual hallucination or pseudo-hallucination of oneself. The cause of autoscopic phenomenon is unknown, but a biological hypothesis is that abnormal, episodic activity in areas of the temporoparietal lobes is involved with the sense of self, perhaps combined with abnormal activity in parts of the visual cortex. It is an uncommon symptom reported in epilepsy, focal lesions/tumours (particularly in the parieto-occipital region), toxic infective states, schizophrenia, depression, anxiety, and dissociative disorders. Near-death experiences often involve an out-of-body experience of viewing one's own body (autoscopy) and overhearing conversations, feelings of peace, hearing a distant voice, entering a dark tunnel, leaving the body behind, meeting dead loved ones, witnessing beings of light, returning to life to complete unfinished business, and deep sadness on leaving this new dimension.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 103.

Anzellotti F et al. Autoscopic phenomena: case report and review of literature. Behavioral and Brain Functions. 2011 Dec;7(1):1-1.

Casey P, Kelly B. Fish's clinical pathology: Signs and symptoms in psychiatry, 4th Edition. 2019. p. 36.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 848.

The correct answer is: Autoscopic hallucination

Question 22

Not answered

Marked out of 1.00

On listening to a joke shared by a friend, a woman loses her muscle tone and collapses suddenly. This is described as

Select one:

- Drop attacks
- Cataplexy
- Syncope
- Catatonia
- Catalepsy

Explanation:

Cataplexy is a sudden loss of muscle tone during wakefulness that is evoked by strong, mainly positive, emotions. It is often seen in narcolepsy. The clinical manifestations are varied, ranging from involuntary eye closure and neck weakness to a subtle buckling of the knees to generalized muscle weakness that causes the patient to collapse. Consciousness and awareness of the environment are preserved throughout the episode.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 104.

Kornum BR et al. Narcolepsy. Nature reviews Disease primers. 2017 Feb 9;3(1):1-9.

The correct answer is: Cataplexy

Question 23

Not answered

Marked out of 1.00

With regard to the psychology of attention, all of the following are correct except

Select one:

- Focused attention is modality specific
- Sustained attention refers to vigilance
- Divided attention is the ability to attend to multiple task demands
- Alternating attention is the highest form of attention
- Shadowing is a test for selective attention

Sohlberg and Mateer proposed a hierarchic model based on the recovering of attention processes of brain-damaged patients when coming out of a coma. 1. Focused attention: The ability to respond discretely to specific visual, auditory or tactile stimuli. This is the most primitive form of attention 2. Sustained attention (vigilance): The ability to maintain a consistent behavioral response during continuous and repetitive activity. 3. Selective attention: The ability to maintain a behavioral or cognitive set in the face of distracting or competing stimuli. Shadowing is a test for selective attention in which simultaneously two different inputs are provided, one to each ear, and the subject is required to 'shadow' or follow one of the inputs. 4. Alternating attention: The ability of mental flexibility that allows individuals to shift their focus of attention and move between tasks having different cognitive requirements. 5. Divided attention: This is the highest level of attention, and it refers to the ability to respond simultaneously to multiple tasks or multiple task demands. In dual-task technique which tests divided attention, the subject is asked to attend to both inputs in an attempt to divide attention. Thus the ability to attend to multiple task demands is tested.

The correct answer is: Alternating attention is the highest form of attention

Question 24

Not answered

Marked out of 1.00

Cognitive theory suggests the following as the underlying factor for anxiety disorders:

Select one:

- Somatisation
- Catastrophisation
- Nihilism
- Depersonalisation
- Personalisation

The term catastrophizing was used by Albert Ellis, the founder of rational-emotional therapy, almost four decades ago. Catastrophizing is related to anxiety (e.g., anxiety is associated with the tendency to overemphasize the probability of a catastrophic outcome and the possible consequences of such an outcome). During the past decade, cognitive models of panic disorder have emphasized the role of catastrophizing in panic attacks (Retrieved from http://journals.lww.com/clinicalpain/Fulltext/2001/03000)

The correct answer is: Catastrophisation

Question 25

Not answered

Marked out of 1.00

Visual or iconic stimuli in immediate memory lasts for

Select one:

- 20-30 minutes
- 0.5 seconds
- 2-3 seconds
- 20-30 seconds
- 0.05 seconds

Explanation:

Iconic memory (or iconic store) is one of the sensory registers, allowing a visual image to persist for about half a second (depending on brightness) after a stimulus has ceased. It enables television, which presents 25 still images per second, each given twice to raise the flicker rate to 50 per second, to convey the illusion of a single continuous image which can be perceived as a moving image.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: 0.5 seconds

Question 26

Not answered

Marked out of 1.00

Which one of the following is an example of non homeostatic motives?

Select one:

- Thirst
- Temperature control
- Achievement
- Hunger
- All of the listed options

There are two types of motivating forces; homeostatic and non homeostatic. Homeostatic motives arise from physiological needs such as thirst, hunger and temperature control and are concerned with basic survival. Non homeostatic motives refer to psychological drives such as curiosity, adventure, achievement, etc.

The correct answer is: Achievement

Question 27

Not answered

Marked out of 1.00

Perceptual constancy can be demonstrated for

Select one:

- Location
- Size
- Brightness constancies
- Colour
- All of the listed options

Perceptual constancy: It refers to the ability to perceive the appearance of objects as constant, even though, their retinal images keep changing. It includes several components: size, shape, colour, location and brightness constancies.

The correct answer is: All of the listed options

Question 28

Not answered

Marked out of 1.00

One of the local newspaper reports that drug addicts carry out muggings. An 83-year-old lady was mugged and the public concluded the attacker was a drug addict. Select the cognitive heuristic in this case.

Select one:

- Framing
- Availability
- Simulation
- Representation
- Anchoring

Explanation:

A heuristic is a rough-and-ready procedure or rule of thumb for making a decision, forming a judgement, or solving a problem without the application of an algorithm or an exhaustive comparison of all available options, and hence without any guarantee of obtaining a correct or optimal result. The availability heuristic is a cognitive heuristic through which the frequency or probability of an event is judged by the number of instances of it that can be readily brought to mind. It can generate biased or incorrect judgements.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015

The correct answer is: Availability

Question 29

Not answered

Marked out of 1.00

In a program aimed at cessation of smoking, a 45-year-old woman is required to smoke many cigarettes in a small booth and over a short period of time. What is the process that is employed in this approach?

Select one:

- Aversive conditioning
- Stimulus control
- Systematic desensitisation
- Variable ratio reinforcement
- Habituation

Explanation:

Aversive smoking has been investigated as a smoking cessation technique that involves rapid smoking in a clinic or laboratory. It consists of smoking rapidly to produce an unpleasant syndrome that becomes a conditioned response to smoking and might help the individual stop smoking. However, existing studies have provided insufficient evidence to determine the efficacy of rapid smoking. Its practice has been mostly abandoned.

Ref: Hajek P, Stead LF. Aversive smoking for smoking cessation. Cochrane Database of Systematic Reviews. 2001(3).

Williams JM et al. Rapid smoking may not be aversive in schizophrenia. Nicotine & Tobacco Research. 2012 Feb 7;15(1):262-6.

The correct answer is: Aversive conditioning

Question 30

Not answered

Marked out of 1.00

A rat in the cage initially got electric shocks whenever goes to a particular area of the cage. Now it stays in electric shock free area. What is the learning process involved?

Select one:

- Avoidance conditioning
- Classical conditioning
- Reciprocal inhibition
- Shaping
- Higher order conditioning

Explanation:

In operant conditioning, avoidance conditioning is a form of negative reinforcement in which an organism avoids an aversive stimulus by making a particular response. Typically, a rat is placed in a long, narrow alleyway with a grid floor that become electrified and delivers a shock a few seconds after the presentation of a conditioned stimulus such as a light, and the rat can avoid the painful stimulation by running to the end of the alley and stepping into a non-electrified goal box in the time interval between the signal and the shock.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015

The correct answer is: Avoidance conditioning

Question 31

Not answered

Marked out of 1.00

A forensic investigator tries to reconstruct the crime scene and encourages visits to scene of crime by witnesses. Which recall phenomenon is replicated in this study?

Select one:

- Blocking
- Failure of prospective memory
- Encoding failure
- State dependent memory
- Context dependent memory

Explanation:

Context-dependent memory is a form of state-dependent memory associated with the context in which information is learned and recalled. In a frequently-cited British experiment on context-dependent memory, lists of words that were learned by divers while they were underwater were best recalled when they were again underwater, and conversely words learned on land were best recalled on land. This phenomenon has been useful particularly in crime detection. Reconstructing the crime scene and visits to scenes of crime has been found to be effective in triggering recall of details of the crime. Similarly, returning to a childhood home village/ town or old school/college brings long forgotten memories flooding back. It is sometimes not even necessary to visit a place to evoke memories but simply imagining the place or situation may be sufficient to retrieve a flood of memories.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Context dependent memory

Question 32

Not answered

Marked out of 1.00

Which of the following is a test that primarily measures sustained attention?

Select one:

- Tower of London test
- Wisconsin card sorting test
- Rey verbal learning test
- Continuous performance test
- Trail making test B

Explanation:

Continuous performance tasks are used to measure individual differences in sustained attention. The task is conceptualised as a measure of both selective and sustained attention, though to a lesser degree it may also gauge aspects of impulsivity. It is based on a simple go-no go design, but many variants of it have been described. Sustained attention is the ability to maintain a consistent focus on some continuous activity or stimuli and is associated with impulsivity. Selective attention is the ability to focus on relevant stimuli and ignore competing stimuli.

Ref: Roebuck H et al. Continuous performance tasks: Not just about sustaining attention. Journal of speech, language, and hearing research. 2016 Jun;59(3):501-10.

The correct answer is: Continuous performance test

Question 33

Not answered

Marked out of 1.00

A person with a fear of heights takes to parachute jumping for a hobby. This is best described as

Select one:

- Counterphobic behaviour
- Reciprocal inhibition
- Desensitisation
- reaction formation
- Projection

Counterphobic behaviour is any behavior that seeks out experiences that are consciously or unconsciously feared in an attempt to master those fears. An example would be parachute jumping for a person with a fear of heights.

The correct answer is: Counterphobic behaviour

Question 34

Not answered

Marked out of 1.00

Fundamental attribution error refers to

Select one:

- Attributing personal factors for one's own behaviours
- Attributing external factors for others' behaviours
- Failure to make attributions for observed behaviours
- Attributing personal factors for others' behaviours
- Attributing external factors for one's own behaviours

Explanation:

Fundamental attribution error is a pervasive tendency to underestimate the importance of external situational pressures and to overestimate the importance of internal motives and dispositions in interpreting the behaviour of others.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Attributing personal factors for others' behaviours

Question 35

Not answered

Marked out of 1.00

According to Allport, the denial of opportunity to the minority group due to the majority group practicing their prejudice is termed as

Select one:

- Avoidance
- Discrimination
- Antilocution
- Physical attack
- Extermination

Allport identified the following processes operating during conflicts between the minority and the majority: 1. Antilocution 2. Avoidance 3. Discrimination: Minority group is discriminated against by denying them opportunity and equality. 4. Physical attack 5. Extermination.

The correct answer is: Discrimination

Question 36

Not answered

Marked out of 1.00

Which of the following is NOT true with regard to research ethics?

Select one:

- Written approval must be obtained from ethics committees
- Advertisements can be used to recruit subjects
- Subjects can be recruited by contacting medical colleagues
- A subject is ethically bound to complete a study after giving written informed consent
- Financial incentives can be offered for subjects to participate in research

Subjects, both healthy controls and patients, are reimbursed for their participation in research. But signing a consent form does not mean that they cannot discontinue the study during its course.

The correct answer is: A subject is ethically bound to complete a study after giving written informed consent

Question 37

Not answered

Marked out of 1.00

Which of the following is true with regard to aetiological theories of suicide?

Select one:

- Sensational media reports of suicide reduce acute risk of suicide
- Serotonin receptor binding is decreased in suicidal patients
- The stressors in the stress-diathesis model can include genetic factors, physical illness or childhood experiences
- Serotonin transporter binding is increased in suicidal patients
- Durkheim proposed social models of suicide

Explanation:

Emile Durkheim, in his book *Suicide*, used survey data purporting to show that egoistic suicide, resulting from feelings of self-reproach and sense of failure, arises from a lack of social cohesion; that altruistic suicide, carried out for the benefit of others, arises from a sense of failure to society; and that anomic suicide, arising from a sense that life is pointless, arises from an absence of social norms that he called anomie.

According to stress-diathesis model, a typical stressor includes the acute worsening of a psychiatric disorder or an acute psychosocial crisis. Pessimism and aggression/impulsivity are components of the diathesis for suicidal behaviour; sex, religion, familial/genetic factors, childhood experiences and various other factors influence this diathesis. In the prefrontal cortex of people who committed suicide, serotonin transporter binding was decreased but serotonin receptor binding was increased. Mann (2003). Nature Reviews Neuroscience 4, 819-828.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Durkheim proposed social models of suicide

Question 38

Not answered

Marked out of 1.00

A patient is has decided to stop alcohol use and is accepting the need for a positive change, though she has not taken any steps yet. According to Prochaska and DiClemente she in the stage of

Select one:

- Precontemplation
- Action
- Preparation
- Maintenance
- Acting out

Explanation:

Six stages of change:

(1) **Precontemplation:** not even considering changing behaviour; does not see the behaviour as a problem, minimizes and denies associated risks, and avoids information to the contrary

(2) **Contemplation:** has become aware that the behaviour is a problem but is ambivalent about changing, and likely sees equal or more benefits than costs from the behaviour

(3) **Preparation:** has made a decision to attempt change

(4) **Action:** has implemented a plan and is changing the behaviour

(5) **Maintenance:** maintaining gains made and attempting to improve areas of life that have been harmed

(6) **Relapse:** return to previous behaviour, but with the possibility of gaining useful strategies to extend the maintenance period on the next attempt

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 575.

The correct answer is: Preparation

Question 39

Not answered

Marked out of 1.00

How many women and girls alive today have been subjected to female genital mutilation?

Select one:

- 1 million
- 10 millions
- 100 millions
- 100, 000
- 10000

Explanation:

At least 200 million girls and women alive today have been cut in 30 countries in Africa, the Middle East, and Asia where female genital mutilation (FGM) is concentrated. FGM involves the partial or total removal of external female genitalia or other injury to the female genital organs for non-medical reasons. The practice has no health benefits for girls and women and is a violation of human rights.

Ref: https://www.who.int/news-room/fact-sheets/detail/female-genital-mutilation

The correct answer is: 100 millions

Question 40

Not answered

Marked out of 1.00

Schizophrenic patients were more likely to relapse if discharged to families exhibiting 'high expressed' emotion for more than:

Select one:

- 55 hours per week
- 35 hours per week
- 15 hours per week
- 24 hours per week
- 45 hours per week

Explanation:

Brown et al in 1962 found that schizophrenic patients were more likely to relapse if discharged to families exhibiting high expressed emotion (hostility, critical comments and emotional over-involvement) especially if they were exposed to these families for more than 35 hours per week.

Expressed emotion (EE) is a construct that reflects potentially problematic aspects of the family environment, particularly for people with psychiatric disorder. EE is a measure of how critical, hostile, or overly involved a family member is toward an identified patient and it is thought to represent relational disturbances and transactional patterns between the family member and that patient. Individuals with schizophrenia appear to be at elevated risk for relapse if they live in family environments characterised by high levels of EE. The original EE-relapse link research involved three major criteria: (1) patients who have schizophrenia, (2) the Camberwell Family Interview used as the method of assessing for EE, and (3) a relapse period of nine to twelve months.

Ref: Weintraub MJ et al. Integrity of literature on expressed emotion and relapse in patients with schizophrenia verified by ap-curve analysis. Family process. 2017 Jun;56(2):436-44.

The correct answer is: 35 hours per week

Question 41

Not answered

Marked out of 1.00

All of the following are potential boundary violations between a therapist and a patient except

Select one:

- A private therapist expecting payment for each session of therapy
- A therapist colludes with a patient against a third party
- A clinician accepts gifts from one patient but not others
- Influencing patient for political causes
- A clinician tries to be impressive often by disclosing personal information

If the therapist expects all patients to pay for each session, then he may not be violating any boundaries by asking a patient do so.

The correct answer is: A private therapist expecting payment for each session of therapy

Question 42

Not answered

Marked out of 1.00

Which of the following is true regarding the association of socioeconomic status (SES) with dementia?

Select one:

- Factors explaining socioeconomic inequalities in dementia risk are well understood
- Low SES is a barrier in facilitating behaviours that might prevent dementia
- Socioeconomic differences in dementia risk are explained by genetic factors
- People with high socioeconomic status are at higher risk for dementia
- There is no association between socioeconomic status and dementia

Explanation:

Differences in dementia risk across the gradient of socioeconomic status (SES) exist, but their determinants are not well understood. Social determinants of health, such as income, education, housing quality, and employment, are associated with disparities in Alzheimer disease and health generally. Low SES has been identified as one of the key barriers in facilitating healthy behaviour in midlife in order to prevent dementia in late-life. Several mechanisms can account for this observation. Low SES has been associated with low health literacy, less access to health resources and proper medical care, and poor health behaviours and health outcomes, some of which are related to increased risk of cognitive decline or dementia.

Ref: Powell WR et al. Association of neighborhood-level disadvantage with Alzheimer disease neuropathology. JAMA network open. 2020 Jun 1;3(6):e207559-.

Deckers K et al. Modifiable risk factors explain socioeconomic inequalities in dementia risk: evidence from a population-based prospective cohort study. Journal of Alzheimer's Disease. 2019 Jan 1;71(2):549-57.

The correct answer is: Low SES is a barrier in facilitating behaviours that might prevent dementia

Question 43

Not answered

Marked out of 1.00

Who proposed theories on cognitive development?

Select one:

- Sullivan
- Ainsworth
- Munroe
- Piaget
- Fromm

Explanation:

Swiss psychologist Jean Piaget created a broad theoretical system for the development of cognitive abilities. He emphasised how children think and acquire knowledge.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 1007.

The correct answer is: Piaget

Question 44

Not answered

Marked out of 1.00

Which of the following is a normal stage of grief, according to Lindemann?

Select one:

- Mourning
- Protest
- Acting Out
- Ambivalence
- Bargaining

The classic work on stages of grief came from Erich Lindemann at Massachusetts General Hospital, who studied 101 bereaved people. He wrote an article, published in 1944, titled 'Symptomology and Management of Acute Grief.' In this article he described a set pattern: After an unexpected death, there is an initial shock that lasts 10-14 days. After the initial shock comes a period of intense sadness, and the grieving person may withdraw from social contact. Next comes anger, as the grieving person seems to 'protest' the unexpected death. Finally, within a year or so, the grief is resolved, and the person returns to normal.

The correct answer is: Protest

Question 45

Not answered

Marked out of 1.00

Which period refers to the autistic phase of Mahler's stages of early development?

Select one:

- 6-36 months
- 0-2 months
- 2-6 months
- 6-12 months
- 12-24 months

Explanation:

Margaret Mahler proposed a theory to describe how young children acquire a sense of identity separate from that of their mothers'. The first stage is 'normal autism' from birth to 2 months. In this stage, periods of sleep outweigh periods of arousal in a state reminiscent of intrauterine life.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 892-893.

The correct answer is: 0-2 months

Question 46

Not answered

Marked out of 1.00

The annual incidence of tardive dyskinesia in patients with chronic schizophrenia who are on any antipsychotic treatment is

Select one:

- 4%
- 3%
- 10%
- 1%
- 20%

Explanation:

The cumulative incidence of tardive dyskinesia is about 4% to 5% annually; the prevalence rate is 20% to 30% (Caroff, 2019). Thus, of 100 patients started on antipsychotics, four would have some manifestations of tardive dyskinesia at the end of 1 year. That rate increases after the second or third year. The risk is lower for second-generation antipsychotics (0.8%, with range=0.0%-1.5% in adults and 5.3% in patients aged 54 years and older) compared to 5.4% in adults treated with haloperidol.

Ref: Caroff SN. Overcoming barriers to effective management of tardive dyskinesia. Neuropsychiatric disease and treatment. 2019;15:785.

The correct answer is: 4%

Question 47

Not answered

Marked out of 1.00

The risk of fetal malformations with sodium valproate is approximately

Select one:

- 10%
- 28%
- 1%
- 70%
- 90%

Explanation:

Valproate is associated with strong evidence for increased risk of congenital and major congenital malformations (as many as 11% of exposed children); cleft lip and/or palate, neural tube defects (risk of spina bifida 1-2%), congenital heart anomalies, radial ray defects, ophthalmological and genitourinary anomalies; developmental delay (delayed walking and talking, memory problems, difficulty with speech and language), low IQ, and autism. This risk increases with increasing doses of valproic acid, especially when the dose is more than 1gm/day.

Ref: McAllister-Williams RH et al. British Association for Psychopharmacology consensus guidance on the use of psychotropic medication preconception, in pregnancy and postpartum 2017. Journal of Psychopharmacology. 2017 May;31(5):519-52.

National Institute for Health and Clinical Excellence. Antenatal and postnatal mental health: clinical management and service guide. Updated edition 2020.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 690-691.

The correct answer is: 10%

Question 48

Not answered

Marked out of 1.00

What percentage of patients on clozapine develop agranulocytosis during the first year of treatment?

Select one:

- 15 percent
- 10 percent
- 5 percent
- 1 percent
- 20 percent

Explanation:

Multiple studies have found that agranulocytosis develops in approximately 0.8% of clozapine-treated patients in the first year.

Ref: Mijovic A, MacCabe JH. Clozapine-induced agranulocytosis. Annals of Hematology. 2020 Aug 20:1-6.

The correct answer is: 1 percent

Question 49

Not answered

Marked out of 1.00

The average age of the start of puberty in boys is

Select one:

- 11-12 years
- 10-11 years
- 8-9 years
- 9-10 years
- 14-15 years

Explanation:

In the UK, the average age for boys to begin puberty is 12 years. For girls, it is 11 years.

Ref: https://www.nhs.uk/live-well/sexual-health/stages-of-puberty-what-happens-to-boys-and-girls/

The correct answer is: 11-12 years

Question 50

Not answered

Marked out of 1.00

The strange-situation study of attachment was carried out on

Select one:

- Children aged 6-9 years
- Infants between 1-6 months
- Newborn babies
- Toddlers
- Children aged 12-24 months

Explanation:

Mary Ainsworth devised the Strange Situation procedure with 12- to 18-month infants, which categorised infant-parent relationships. The procedure involves separations and reunions with caregivers and observes response. It has three phases with the infant's reactions being noted in each of the phases: 1. Reaction to a stranger in the presence of the mother 2. Reaction when left alone with a stranger 3. Reaction when reunited with the mother.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 658.

The correct answer is: Children aged 12-24 months

Question 51

Not answered

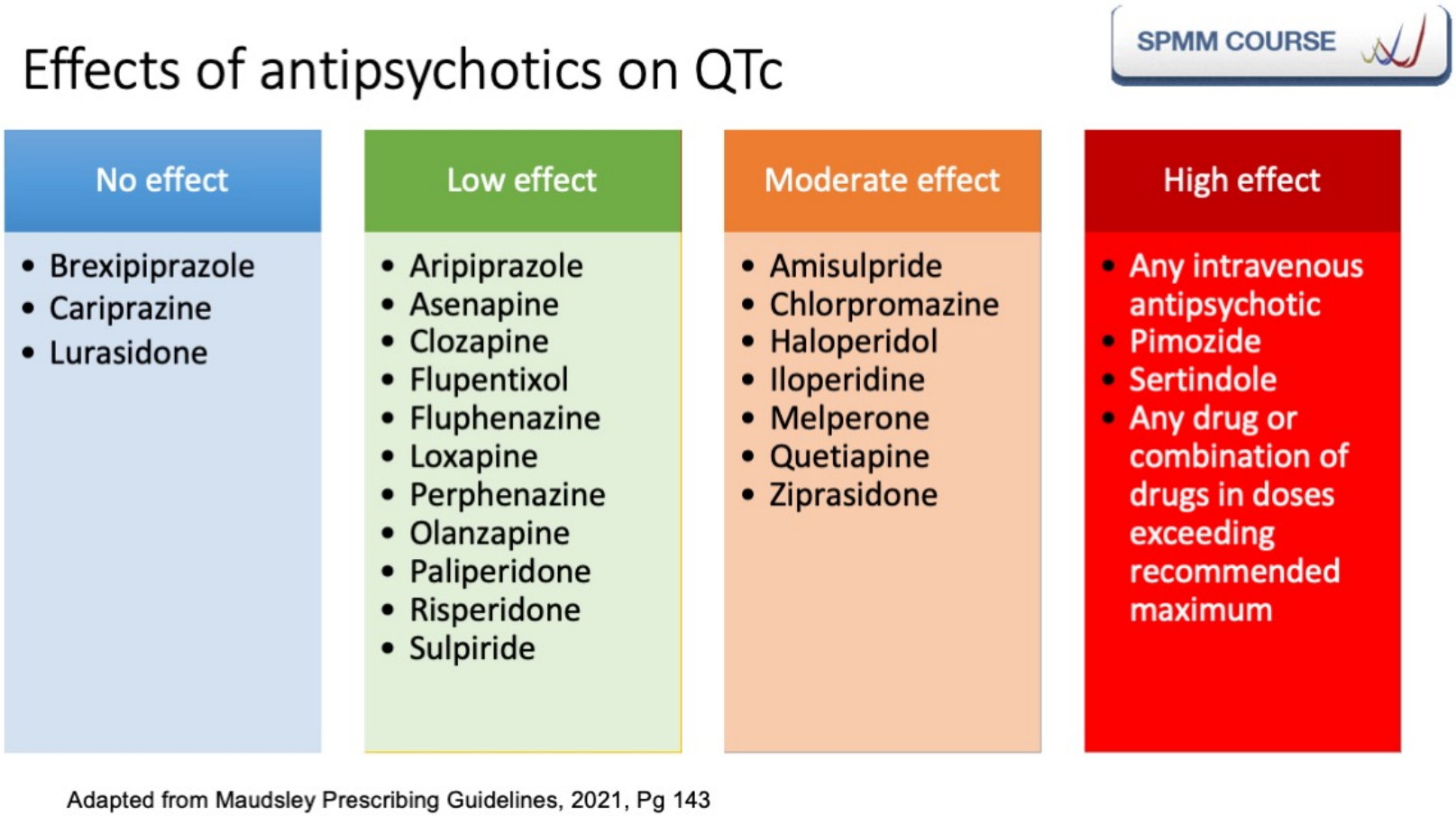
Marked out of 1.00

Which of the following drugs is currently thought to cause least QT prolongation of all antipsychotics?

Select one:

- Olanzapine
- Quetiapine
- Thioridazine
- Clozapine
- Lurasidone

Effects of antipsychotics on QTc:



Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 143.

The correct answer is: Lurasidone

Question 52

Not answered

Marked out of 1.00

According to the CATIE study of patients with schizophrenia, the number of patients meeting criteria for metabolic syndrome ranges between what percentages?

Select one:

- 11-25%
- 71-85%
- 56-70%
- 26-40%
- 41-55%

Explanation:

Depending on the definition of 'metabolic syndrome' (MS), its prevalence at baseline in CATIE was 41-43% in males and 52-54% in females.

Ref: McEvoy JP et al. Prevalence of the metabolic syndrome in patients with schizophrenia: baseline results from the Clinical Antipsychotic Trials of Intervention Effectiveness (CATIE) schizophrenia trial and comparison with national estimates from NHANES III. Schizophrenia research. 2005 Dec 1;80(1):19-32.

The correct answer is: 41-55%

Question 53

Not answered

Marked out of 1.00

The upper normal QTc level in female subjects is

Select one:

- 440 ms
- 450 ms
- 500 ms
- 470 ms
- 420 ms

Explanation:

The upper limit of normal QTc interval is 440ms for men and 470ms for women. Rather strong evidence links QTc values over 500ms to a clearly increased risk of arrhythmia.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 142.

The correct answer is: 470 ms

Question 54

Not answered

Marked out of 1.00

A 44-year-old woman on clozapine has gained 15Kgs in 3 months after starting the treatment. This antipsychotic induced weight gain is most likely to be due to;

Select one:

- Hypoprolactinaemia
- Leptin sensitisation
- 5-HT2C antagonism
- M4 blockade
- H2 antagonism

Explanation:

The weight gain induced by clozapine may be partially associated with its potent blockade of both H1 histamine and 5HT2C receptors. Agonists at the 5HT2 family of receptors cause hypophagia and can cause weight loss in animals and human beings. In rats, 5HT2C antagonists cause an increase in feeding. 5HT2C knockout mice are overweight and exhibit increased feeding.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 225.

The correct answer is: 5-HT2C antagonism

Question 55

Not answered

Marked out of 1.00

Which of the following is true with regard to acetylcholine (Ach) ?

Select one:

- Acetylcholinergic projections are more predominant than serotonergic projections in the cortex
- Tricyclics increase acetylcholine levels
- Antidementia drugs reduce acetylcholine concentration
- Acetylcholine agonists are useful in mild to moderate dementia of Alzheimer's type
- Acetylcholine is synthesised by acetyl cholinesterase

The distribution of cholinergic innervation in the human cerebral cortex reveals that all cytoarchitectonic regions and layers of the cerebral cortex display a dense cholinergic innervation The density of cholinergic axons is higher in the more superficial layers (layers I and II and the upper parts of layer III) of the cerebral cortex. Distinct patterns of lamination exist in individual cytoarchitectonic regions (Structure and Function of Cholinergic Pathways. http://www.acnp.org/g4/GN401000012/CH012.html (accessed April 13, 2015)).

The correct answer is: Acetylcholinergic projections are more predominant than serotonergic projections in the cortex

Question 56

Not answered

Marked out of 1.00

Which of the following antidepressants is least likely to be associated with sexual dysfunction in male depressed patients?

Select one:

- Agomelatine
- Trazodone
- Venlafaxine
- Fluoxetine
- Amitriptyline

Explanation:

Lower risk antidepressants include agomelatine, bupropion, mirtazapine, vilazodone, vortioxetine, and moclobemide. Of these, agomelatine, bupropion, and vortioxetine have the best evidence supporting a more favourable sexual adverse-effect profile. Rates of sexual dysfunction with agomelatine may be similar to placebo.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 405-406.

The correct answer is: Agomelatine

Question 57

Not answered

Marked out of 1.00

Mr. X is taking trazodone for his depressive symptoms. He complains of excessive sedation. This effect is most likely to be due to the blockade of which of the following receptors?

Select one:

- Nicotinic
- Beta adrenergic
- Alpha adrenergic
- Muscarinic
- Dopaminergic

Explanation:

Trazodone blocks the serotonin transporter at antidepressant doses. It also has serotonin 2A and 2C, H1 histamine, and alpha-1-adrenergic antagonist properties, which can make it very sedating. At low doses, trazodone does not adequately block serotonin reuptake but retains its other properties.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 535.

The correct answer is: Alpha adrenergic

Question 58

Not answered

Marked out of 1.00

Mr.Z is a 26 year old gentleman who was diagnosed with paranoid schizophrenia 4 years ago and has had 6 hospitalisations over last 3 year period. Most of his relapses have been associated with non-adherence to medication. Which of the following interventions are most likely to improve compliance for this patient?

Select one:

- Adherence therapy
- Addressing patient-specific barriers
- Supportive psychotherapy
- Family therapy
- Psychoeducational intervention

Explanation:

Current guidelines recommend that interventions to improve medication adherence should be patient specific, in that they should target the barriers to achieving adherence as perceived or noted by the clinical team to be present in that patient (Barnes et al., 2020).

Of various other psychosocial approaches, Adherence Therapy, a brief psychological intervention based on aspects of motivational interviewing and CBT, is the best studied. While this intervention showed some early promise for improving insight, adherence, attitudes towards medication, and rehospitalisation rates in an inpatient sample of patients with schizophrenia, further studies failed to replicate these positive findings for improving adherence. A well-conducted, adequately powered RCT found adherence therapy to be ineffective in improving adherence, psychotic symptoms, or quality of life in people with schizophrenia. Meta-analysis of adjunct use of adherence therapy support its use in reducing symptom burden but surprisingly, no significant changes in adherence per se (Gray et al, 2016).

Similarly, psychoeducation alone or family therapy alone are also ineffective. There is no evidence based data available for supportive psychotherapy and insight orientated psychotherapy for improving medication adherence (Curto et al., 2021).

Ref: Barnes TR et al. Evidence-based guidelines for the pharmacological treatment of schizophrenia: updated recommendations from the British Association for Psychopharmacology. Journal of Psychopharmacology. 2020 Jan;34(1):3-78.

Gray R, et al. Is adherence therapy an effective adjunct treatment for patients with schizophrenia spectrum disorders? A systematic review and meta-analysis. BMC psychiatry. 2016 Dec;16(1):1-2.

Curto M, Fazio F, Ulivieri M, Navari S, Lionetto L, Baldessarini RJ. Improving adherence to pharmacological treatment for schizophrenia: a systematic assessment. Expert Opinion on Pharmacotherapy. 2021 Jun 13;22(9):1143-55.

The correct answer is: Addressing patient-specific barriers

Question 59

Not answered

Marked out of 1.00

Choose the most accurate statement about cannabis

Select one:

- Cannabis commonly causes physical dependence
- 9-delta-tetrahydrocannabinol is the active substance
- Fatal overdose is reported in several countries
- Urine tests usually remain positive for up to a week after regular use has ceased
- Anandamide is a specific cannabinoid receptor and a naturally occurring antagonist at this receptor

Explanation:

Cannabis is a mixture of hundreds of chemical and over 100 alkaloid cannabinoids. The most important of these are tetrahydrocannabinol (THC) and cannabidiol (CBD). THC interacts with cannabinoid 1 and 2 (CB 1 and CB2) receptors and has psychoactive properties. CBD is an isomer of THC and relatively inactive at CB1 and CB2 receptors. CBD does not have psychoactive properties and its mechanism of action is unknown. Cannabis comes in various mixtures of THC and CBD.

Anandamide is a specific cannabinoid receptor and a naturally occurring agonist at this receptor. Cannabis is not usually associated with physical dependence. Urine tests usually remain positive for up to 4weeks after regular use has ceased. Unlike most other drugs of abuse, there are no reports of fatal overdose from cannabis.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 563.

The correct answer is: 9-delta-tetrahydrocannabinol is the active substance

Question 60

Not answered

Marked out of 1.00

A 66 year old man taking antidepressant therapy develops dizziness every morning. He gives a history of palpitations accompanied by the dizziness. There is no history of vertigo or falls but the dizziness improves spontaneously on reclining. Which of the following receptor blockade can explain the above symptoms?

Select one:

- Alpha 1 adrenergic
- Cholinergic nicotinic
- Cholinergic muscarinic
- Alpha 2 adrenergic
- Beta adrenergic

Explanation:

Alpha-1 adrenergic blockade can cause dizziness, sedation, and hypotension.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021.

The correct answer is: Alpha 1 adrenergic

Question 61

Not answered

Marked out of 1.00

Carlsson is associated with which of the following?

Select one:

- learned helplessness hypothesis of depression
- hairline triggers in borderline personality
- endocrine hypothesis of depression
- kindling hypothesis of depression
- dopamine hypothesis of schizophrenia

Explanation:

Arvid Carlsson, a Swedish pharmacologist, is best known for his contributions on the neurotransmitter, dopamine, for which he won the Nobel Prize in 2000 for Medicine/Physiology. The co-recipients were Eric Kendel and Paul Greengard. In 1957, Carlsson showed that dopamine was a neurotransmitter in the brain and not just a precursor of norepinephrine, which was the prevailing view at the time. He was also instrumental in developing the 'dopamine theory of schizophrenia' and the role of dopamine in the development of extrapyramidal side effects of antipsychotic medications.

Ref: Yeragani VK et al. Arvid Carlsson, and the story of dopamine. Indian journal of psychiatry. 2010 Jan;52(1):87.

The correct answer is: dopamine hypothesis of schizophrenia

Question 62

Not answered

Marked out of 1.00

The risk factor predicting poor response to lithium prophylaxis in bipolar illness is

Select one:

- Dominant personality
- Social support
- High social status
- High age of illness onset
- High number of previous hospitalizations

Predictors of poor response to lithium in bipolar prophylaxis include High number of previous hospitalization, stress, high expressed emotion, and high number of life events, episodic patterns of depression- mania interval, continuous cycling, and unemployment and neurotic personality.

The correct answer is: High number of previous hospitalizations

Question 63

Not answered

Marked out of 1.00

Which of the following antipsychotics is recommended for mixed affective states in bipolar disorder?

Select one:

- perphenazine
- amisulpride
- sulpride
- clozapine
- olanzapine

Explanation:

According to current guidelines, if a patient with bipolar disorder presents in a mixed state, consider stopping the antidepressant (if the patient is taking one) and offer an antipsychotic (haloperidol, olanzapine, quetiapine, or risperidone).

Ref: National Institute for Health and Clinical Excellence. The NICE guideline on the assessment and management of bipolar disorder in adults, children and young people in primary and secondary care. Updated edition 2020.

The correct answer is: olanzapine

Question 64

Not answered

Marked out of 1.00

A 36-year-old cocaine user notices that the longer he uses the drug, the more of it he requires to achieve the same effect. This is best described as

Select one:

- Addiction
- Dependence
- Tolerance
- Sensitisation
- Compliance

Explanation:

The development of tolerance is marked by a need, over time, to use increased doses of a drug for it to achieve the same effect.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 604.

The correct answer is: Tolerance

Question 65

Not answered

Marked out of 1.00

Which of the following is true with regard to nicotinic receptors?

Select one:

- Blockade causes problems with fine motor movements
- Mediate anti-parkinsonian effects of amantadine
- All nicotinic receptors are excitatory
- Mediate sympathomimetic actions of dopamine
- Anticholinergic effects of antipsychotics are mediated by nicotinic receptors

Explanation:

There are numerous receptors for acetylcholine. The major subtypes are nicotinic and muscarinic subtypes of cholinergic receptors. Nicotinic receptors are all ligand-gated, rapid-onset, and excitatory ion channels blocked by curare. Muscarinic receptors, by contrast, are G-protein-linked, can be excitatory or inhibitory, and many are blocked by atropine, scopolamine, and other well-known so-called 'anticholinergics'.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 506.

The correct answer is: All nicotinic receptors are excitatory

Question 66

Not answered

Marked out of 1.00

Agonistic effect of fluoxetine on which of the following receptors results in anorgasmia?

Select one:

- Alpha-1 adrenergic receptors
- 5HT-1A stimulation
- D-2 receptors
- 5HT-2C stimulation
- Mu opioid receptors

Explanation:

Psychotropics affect sexual functioning through the dopamine and 5-HT systems. Antipsychotics blocking the D2 receptors and antidepressants increasing 5-HT transmission via 5-HT2 receptors affect sexual functioning. D2 blockade (direct or indirectly via 5HT2A stimulation) can cause apathy and reduced libido. Agonistic stimulation of 5-HT2A and 5-HT2C inhibits the spinal reflexes of orgasm and ejaculation leading to anorgasmia and delayed ejaculation, as seen with the use of SSRIs in some patients. 5-HT2A antagonists (mirtazapine) and 5-HT1A autoreceptor agonists (e.g., buspirone) reverse serotonergic inhibition and thus have a more favourable sexual side effect profile.

Ref: Lew-Starowicz, M. and Giraldi, A., 2021. Impact of Psychotropic Medications on Sexual Functioning. In Psychiatry and Sexual Medicine (pp. 353-371). Springer, Cham.

The correct answer is: 5HT-2C stimulation

Question 67

Not answered

Marked out of 1.00

SSRIs, on stimulating which receptors, will induce sleep disturbances?

Select one:

- 5-HT2A
- 5-HT3
- 5-HT1A
- H-1
- 5-HT1D

5-HT2A stimulation by SSRIs will lead to insomnia

The correct answer is: 5-HT2A

Question 68

Not answered

Marked out of 1.00

Aripiprazole has an antagonistic effect on which of the following receptors?

Select one:

- D2 receptors
- 5-HT3 receptors
- 5-HT2A receptors
- 5-HT1A receptors
- D1 receptors

Explanation:

Aripiprazole is a D2 receptor partial agonist. It is also a partial agonist at 5-HT1A receptors, a high affinity antagonist at 5-HT2A receptors, and a low-/moderate affinity antagonist at H1 and alpha-1 receptors.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 212.

The correct answer is: 5-HT2A receptors

Question 69

Not answered

Marked out of 1.00

Which of the following is a noncompetitive NMDA blocker?

Select one:

- Lamotrigine
- Bupropion
- Acamprosate
- Disulfiram
- Varenicline

Explanation:

Acamprosate is a synthetic taurine analogue that acts as a functional glutaminergic NMDA antagonist and also increases GABA-ergic function. Thus, it interacts both with the glutamate system to inhibit it, and with the GABA system to enhance it, a bit like a form of 'artificial alcohol'.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 463.

Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 556.

The correct answer is: Acamprosate

Question 70

Not answered

Marked out of 1.00

The number of half-lives normally taken by drugs to reach steady plasma levels is;

Select one:

- 7
- 9
- 2
- 3
- 5

Explanation:

It usually takes 4-5 drug half-lives to reach steady-state levels.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 833.

The correct answer is: 5

Question 71

Not answered

Marked out of 1.00

Which of the following is correct with regard to benzodiazepine metabolism?

Select one:

- Desmethyldiazepam is an inactive metabolite
- Diazepam achieves nearly 10 times greater plasma level in the elderly
- Diazepam is around 95% protein bound
- Duration of action depends primarily on elimination half-life
- Elimination half-life is reduced in the elderly

Explanation:

Clinically, half-life alone does not necessarily determine the duration of action for most benzodiazepines. The fact that all benzodiazepines are lipid-soluble to varying degrees means that benzodiazepines and their active metabolites bind to plasma proteins. The extent of this binding is proportional to their lipid solubility. The amount of protein binding varies from 70-99% (approximately 95% for diazepam). Distribution, onset, and termination of action after a single dose are thus determined mainly by benzodiazepine lipid solubility, not elimination half-life. Diazepam is oxidised to desmethyldiazepam, which, in turn, is hydroxylated to produce oxazepam. These active metabolites undergo glucuronidation to inactive metabolites. The elimination half-life of diazepam is increased by 100% in elderly (around 90 hrs) as clearance is reduced.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 670.

Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021.

The correct answer is: Diazepam is around 95% protein bound

Question 72

Not answered

Marked out of 1.00

Mr Y is a 27 year old gentleman who has recently been started on lithium for treatment of his acute phase of his manic episode. What is the therapeutic blood lithium level you would aim for?

Select one:

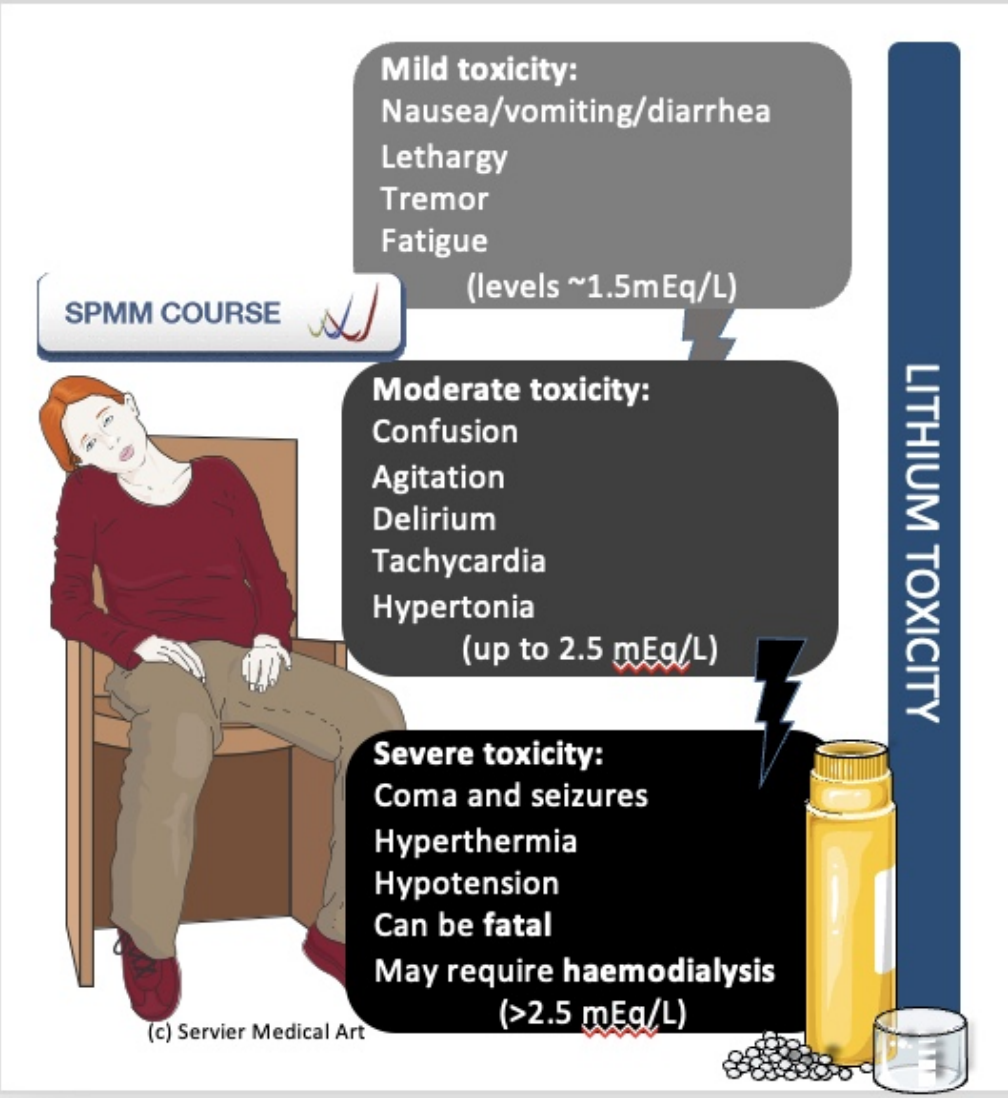
- 0.6 - 0.8 M Eq/ L
- 0.4 - 0.5 M Eq/ L
- 0.2 - 0.4 M Eq/ L
- 1.5 - 2.0 M Eq/ L
- 0.8 - 1.0 M Eq/ L

Explanation:

Lithium is currently the drug of choice for treatment of the acute phase of a manic episode as well as for prevention of further episodes of bipolar mood disorder. The treatment is closely monitored by repeated blood levels as the difference between therapeutic and lethal blood levels is not very wide.

Lithium is effective for the treatment of acute mania, at a plasma level of 0.8-1.0mmol/L. For maintenance treatment, aim for the highest possible tolerable lithium plasma level in the range of 0.6-0.8 mmol/L. A blood lithium level of more than 2 mmol per litre may be associated with toxicity while a level of more than 2.5 may be lethal.

Note that lithium is a monovalent cation; so the therapeutic range expressed in mEq/L is identical to mmol/L



Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 247-248.

Goodwin GM et al. Evidence-based guidelines for treating bipolar disorder: revised third edition recommendations from the British Association for Psychopharmacology. Journal of Psychopharmacology. 2016 Jun;30(6):495-553.

The correct answer is: 0.8 - 1.0 M Eq/ L

Question 73

Not answered

Marked out of 1.00

The minimum effective dose of Aripiprazole in treating a psychotic episode is

Select one:

- 20mg
- 15mg
- 10mg
- 25mg
- 5mg

Explanation:

Antipsychotic	Minimum effective dose/day in first episode schizophrenia	Minimum effective dose/day in multi-episode schizophrenia
Chlorpromazine	200mg	300mg
Haloperidol	2mg	4mg
Sulpiride	400mg	800mg
Trifluoperazine	10mg	15mg
Amisulpride	300mg	400mg
Aripiprazole	10mg	10mg
Asenapine	10mg	10mg
Blonanserin	Not known	8mg
Brexipiprazole	2mg	2mg
Cariprazine	1.5mg	1.5mg
Iloperidone	4mg	8mg
Lumateperone	Not known	42mg
Lurasidone	40mg HCl/37mg base	40mg HCl/37 mg base
Olanzapine	5mg	7.5mg
Quetiapine	150mg (but higher doses often used)	300mg
Risperidone	2mg	4mg
Ziprasidone	40mg	80mg

Adapted from Maudsley Guidelines 2021, p.9.

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Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 9.

The correct answer is: 10mg

Question 74

Not answered

Marked out of 1.00

The minimum duration of gap recommended between stopping venlafaxine and starting duloxetine is

Select one:

- No gap necessary
- 2 months
- 2 days
- 1 day
- 1 week

Explanation:

A direct switch can be made when switching from venlafaxine to duloxetine.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021, p. 341.

The correct answer is: No gap necessary

Question 75

Not answered

Marked out of 1.00

The rate-limiting enzyme in the metabolism of alcohol is;

Select one:

- Alcohol dehydrogenase
- Acetyl co A dehydrgenase
- Aldehyde dehydrogenase
- Acetic acid
- Acetyl dehydrogenase

The major site of alcohol metabolism is the liver. Alcohol dehydrogenase is the rate-limiting enzyme in the metabolism of alcohol which makes it impossible to speed up alcohol metabolism.

The correct answer is: Alcohol dehydrogenase

Question 76

Not answered

Marked out of 1.00

Which of the following is a localising sign for frontal lesions?

Select one:

- Postural headache
- Ptosis
- Nystagmus
- Olfactory disturbances
- Myoclonus

The most classic presentation of frontal lobe dysfunction is an olfactory groove meningioma characterized by anosmia, loss of inhibition, memory impairment, headaches, and visual symptoms. But anosmia itself is not a good localising sign as often the cause of olfactory disturbance is peripheral - related to nostrils (Frontal Lobe Syndromes Clinical Presentation, http://emedicine.medscape.com/article/1135866-clinical (accessed April 13, 2015).).

The correct answer is: Olfactory disturbances

Question 77

Not answered

Marked out of 1.00

National adult reading test is used to assess

Select one:

- Visuospatial ability
- Premorbid intelligence
- Attentional bias
- Nonverbal intelligence
- Frontal lobe functions

Explanation:

First published in 1982, the 50-item National Adult Reading Test (NART) is a widely adopted method for estimating premorbid intelligence both for clinical and research purposes. The validity of NART as a measure of premorbid ability rests upon the assumptions that reading ability (of irregular words) is relatively independent of brain damage, and that it is a strong predictor of intelligence in the normal population.

Ref: Bright P et al. The National Adult Reading Test: restandardisation against the Wechsler adult intelligence scale—fourth edition. Neuropsychological Rehabilitation. 2018 Aug 18;28(6):1019-27.

The correct answer is: Premorbid intelligence

Question 78

Not answered

Marked out of 1.00

Which of the following drugs have a half-life of 3 days?

Select one:

- Carbamazepine
- Aripiprazole
- Quetiapine
- Reboxetine
- Fluoxetine

Explanation:

The mean elimination half-life is 75 hours for aripiprazole and 94 hours for the major metabolite dehydro-aripiprazole.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 57.

The correct answer is: Aripiprazole

Question 79

Not answered

Marked out of 1.00

Which of the following drugs is least dependent on hepatic metabolism?

Select one:

- Chlorpromazine
- Clozapine
- Olanzapine
- Amisulpride
- Risperidone

Explanation:

Gabapentin, sulpiride, amisulpride, and lithium all undergo no or minimal hepatic metabolism.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 724.

The correct answer is: Amisulpride

Question 80

Not answered

Marked out of 1.00

A 34-year-old man unable to remember any of the events that happened at the hospital during the time of his admission for a head injury even after two years after recovery. He has no other memory problems. This is

Select one:

- Confabulation
- Anterograde amnesia
- Retrograde amnesia
- Episodic memory loss
- Dissociative memory loss

Explanation:

Anterograde amnesia is the period of amnesia between an event (e.g. head injury) and the resumption of continuous memory. The length of anterograde amnesia is correlated with the extent of brain injury.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 102.

The correct answer is: Anterograde amnesia

Question 81

Not answered

Marked out of 1.00

A 70-year-old man presents with sudden onset of right arm weakness following severe headache and inability to understand speech. His speech problem is accurately described as

Select one:

- Aphasia
- Alexia
- Aphemia
- Agraphia
- Amusia

Explanation:

Aphasia is an impairment of expression or comprehension of language caused by injury or disease in the language centres of the brain. This case is receptive aphasia (Wernicke's area damage).

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Aphasia

Question 82

Not answered

Marked out of 1.00

On routine examination of eyes, accommodation reflexes were present but pupils were small irregular with absent light reflexes. This type of pupil is called

Select one:

- Marcus Gunn pupil
- Senile pupil
- Anosognosia
- Kayser-Fleischer pupil
- Argyll-Robertson pupil

Explanation:

An Argyll-Robertson pupil is a pupil of the eye that manifests accommodation by constricting when watching a finger that is brought close to the face but does not respond to light by constricting, a neurological sign of damage to the midbrain relays of the oculomotor nerve. Douglas Argyll Robertson's (1837-1909) conducted experimental work with physostigmine in 1863. He observed patients with spinal cord disease who had lost the response to light even though accommodation to near objects was normal. This phenomenon, known subsequently as the Argyll Robertson pupil, came to be considered pathognomonic of tabes dorsalis, general paresis and neurovascular syphilis.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Argyll-Robertson pupil

Question 83

Not answered

Marked out of 1.00

Abnormalities in Stroop test suggests

Select one:

- Short term memory disturbance
- Damage to occipital cortex
- Attentional bias
- Cortical blindness
- Lesion of optic chiasma

Explanation:

The Stroop effect is the delay and disruption in naming the colours of words printed in non-matching coloured ink, as when the word 'red' is printed in blue ink, the word 'blue' in green ink, and so on. To perform the task it is necessary to ignore the meanings of the printed words and to respond only to the colours in which they are printed, but with experienced adult readers automatic processing tends to occur, causing interference in the processing of information and a significant increase in the time required to name the colours, compared with the time taken to name the colours of meaningless strings of letters.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015

The correct answer is: Attentional bias

Question 84

Not answered

Marked out of 1.00

Which of the following instruments can be used to screen for cognitive deficits after head injury?

Select one:

- Bender visual motor gestalt test
- National adult reading test
- Glasgow coma scale
- Weschler adult intelligence scale
- Raven's progressive matrices

Explanation:

The Bender Visual-Motor Gestalt Test is designed to help in the diagnosis of loss of function and organic brain damage in children and adults. It consists of nine geometrical figures that are copied by the respondent, the drawings being evaluated according to the overall quality of the reproductions, their organisation in relation to the Gestalt grouping laws, and the errors made. Individuals who have suffered a traumatic brain injury may be given the Bender-Gestalt as part of a battery of neuropsychological measures or tests.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015

The correct answer is: Bender visual motor gestalt test

Question 85

Not answered

Marked out of 1.00

A 64-year-old man has behavioural changes with irritability. He is found wandering in the streets near his home. On neurological examination, there is evidence of mild aphasia and cognitive dysfunction, but motor function is preserved. He seems to have no awareness regarding his difficulties and has a silly, jocular affect. Which of the following is most likely to be seen in CT/MRI scan?

Select one:

- Dilated ventricles without cerebral atrophy
- Predominant occipito parietal atrophy
- Diffuse cortical atrophy with apparently dilated ventricles
- Bilateral frontal atrophy
- Space occupying lesion obstructing third ventricle

The presence of reduced verbal fluency, loss of insight and having a silly affect points towards frontal lesions.

The correct answer is: Bilateral frontal atrophy

Question 86

Not answered

Marked out of 1.00

The gene implicated in Huntington disease is present on the chromosome

Select one:

- 21
- 14
- 7
- 4
- 1

Explanation:

Huntington disease results from expansion of a CAG trinucleotide repeat in the HTT (huntingtin) gene on the short arm of chromosome 4. Alleles with 40 or more trinucleotide repeats are fully penetrant.

Ref: Bachoud-Lévi AC et al. International Guidelines for the treatment of Huntington's Disease. Frontiers in neurology. 2019;10:710.

The correct answer is: 4

Question 87

Not answered

Marked out of 1.00

Which of the following is associated with narcolepsy?

Select one:

- Waxy flexibility
- Unrefreshing sleep attacks
- 2% prevalence
- Prolonged REM latency
- HLA-DR2

Explanation:

Narcolepsy is a chronic disorder that is primarily characterised by excessive daytime sleepiness and, in most individuals, cataplexy (NOT catalepsy / waxy flexibility - these are catatonic features). Narcolepsy is the most common neurological cause of hypersomnia with an estimated prevalence of 0.20-0.40 per 1000 in the general population. The absence of the hypocretin peptides (also known as orexins; which are neuropeptides with a role in the regulation of sleep and wakefulness) is a fundamental marker in most patients with narcolepsy with cataplexy. The hypocretin deficiency in narcolepsy is thought to result in an unstable switch mechanism, with uncontrolled daytime sleep attacks (which may be refreshing, as they are REM attacks). There is a strong association between HLA-DR2 haplotypes coded on chromosome 6 and narcolepsy - HLA-DQB1*0602 and DQA1*0102 are found in up to 85-95% of individuals, compared with 12-38% in the general population.

Investigation	Findings in narcolepsy
Polysomnography (sleep EEG and Multiple Sleep Latency Test)	Sleep-onset REM period is highly-specific (25-50% of cases) Increased N1 (light sleep) and repeated awakenings
CSF hypocretin-1 levels	Levels <110pg/mL are highly specific and sensitive for narcolepsy with cataplexy (In 10%, levels may be normal or even high)
HLA typing	Strong association with HLA-DR2 haplotypes coded on chromosome 6 HLA DQB1*0602 and DQA1*0102 are found in up to 85-95% of individuals (compared with 12-38% in the general population)
Imaging	MRI useful to exclude some rare causes of secondary narcolepsy (abnormalities of the brainstem and diencephalon)
	Adapted from Oxford Handbook of Psychiatry, 2019. p. 451.

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Ref: Kornum BR et al. Narcolepsy. Nature reviews Disease primers. 2017 Feb 9;3(1):1-9.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 450-451.

The correct answer is: HLA-DR2

Question 88

Not answered

Marked out of 1.00

Mr. Y has a well-established diagnosis of bipolar affective disorder. His younger brother, who is also his main carer, is aware that this disorder has a strong genetic loading and is worried. What is the prevalence rate in first-degree relatives of patients with bipolar affective disorder?

Select one:

- 10-15%
- 20-40%
- 1-2%
- 5-7%
- 60-70%

Explanation:

Bipolar disorder has a multifactorial aetiology. Both family studies and twin studies have demonstrated a strong genetic loading for it.

First-degree relatives of those with bipolar disorder are 7 times more likely to develop bipolar disorder than the general population (i.e. 10-15% risk). Monozygotic twins have a concordance of 33-90%. Dizygotic twins have a concordance of approximately 23%.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. pp. 332-333.

Rowland TA, Marwaha S. Epidemiology and risk factors for bipolar disorder. Therapeutic advances in psychopharmacology. 2018 Sep;8(9):251-69.

The correct answer is: 10-15%

Question 89

Not answered

Marked out of 1.00

What is the concordance rate of schizophrenia in monozygotic twins?

Select one:

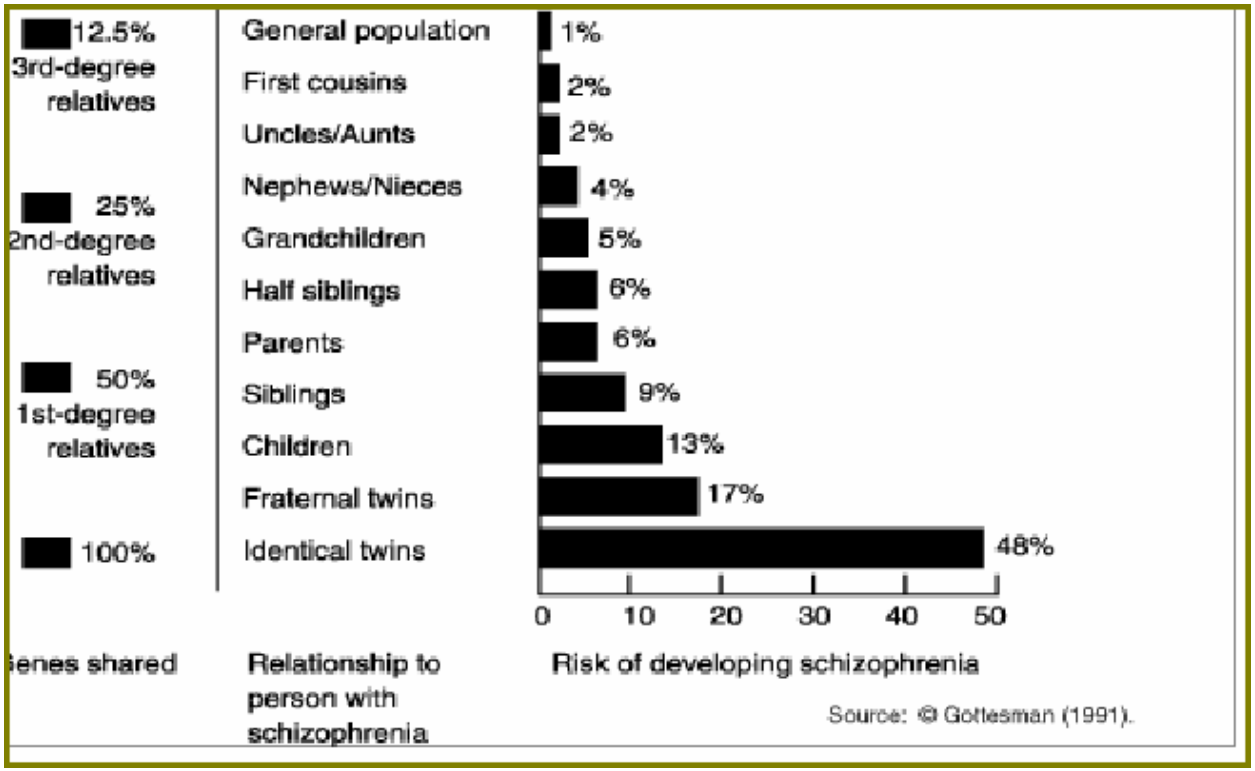
- 22%
- 48%
- 18%
- 14%
- 8%

Explanation:

Genetic factors account for most of the liability to schizophrenia. Estimated heritability ranges from 64% (pedigree studies) to 81% (twin studies). Most studies suggest that the concordance rates are 44-50% for monozygotic twins and 10-15% for dizygotic twins, i.e., a three times higher concordance in monozygotic than in dizygotic twins. The risk of developing schizophrenia when one has an affected identical twin is approximately 46-48%.

Ref: Jauhar, S et al. Schizophrenia (Seminar). The Lancet, 2022: 399, 10323, pg473-486

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. pp. 190-191.



The correct answer is: 48%

Question 90

Not answered

Marked out of 1.00

Nonfluent aphasia is associated with lesions in or near which of the following brain regions?

Select one:

- Parietal operculum
- Anterior cingulate
- Superior temporal gyrus
- Insula
- Broca area

Explanation:

Broca aphasia (also called non-fluent or expressive aphasia) had traditionally been characterised by non-fluent speech and impaired repetition but intact auditory comprehension. It has long been thought to be associated with damage to Broca area (i.e. left inferior frontal convolution) or Brodmann area 44. Temporal and parietal lobes are usually not impaired. Therefore, receptive language and comprehension are not usually impaired in nonfluent aphasia.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 48.

The correct answer is: Broca area

Question 91

Not answered

Marked out of 1.00

Bobby is an 8-year-old boy who has consistently shown callous-unemotional traits (low empathy, callousness, and low interpersonal emotions). What is the heritability (genetic contribution) of these traits?

Select one:

- 5-10%
- 70-80%
- 1-2%
- 10-20%
- 40-60%

Explanation:

Twin studies suggest that callous-unemotional traits in middle childhood are highly heritable. Heritability estimates range from 0.45 to 0.67.

Ref: Hyde LW et al. Heritable and nonheritable pathways to early callous-unemotional behaviors. American Journal of Psychiatry. 2016 Sep 1;173(9):903-10.

Moore AA et al. The Inventory of Callous-Unemotional Traits (ICU) in children: reliability and heritability. Behavior genetics. 2017 Mar 1;47(2):141-51.

The correct answer is: 40-60%

Question 92

Not answered

Marked out of 1.00

Which of the following substances is found in reduced levels in patients with high impulsive suicidality?

Select one:

- CYP4503A4
- COMT
- 5-HIAA
- BDNF
- Neuregulin

Explanation:

Much of the clinical data relating 5-HT and impulsivity have been provided by investigations of suicide. Early studies reported lower CSF and plasma 5-HIAA levels as well as blunted prolactin response to fenfluramine in both suicide attempters and completers, as well as lower brain 5-HT, 5-HT transporter, and abnormal 5-HT receptor binding at post-mortem in suicide completers. Research into individuals with antisocial personality disorder categorised according to whether their violent behaviour was aggressive or non-aggressive has revealed similar results, with impulsive aggressive individuals reported to have lower levels of CSF 5-HIAA and blunted prolactin response to fenfluramine.

Ref: Dalley JW, Roiser JP. Dopamine, serotonin and impulsivity. Neuroscience. 2012 Jul 26;215:42-58.

The correct answer is: 5-HIAA

Question 93

Not answered

Marked out of 1.00

Which of the following types of chromosomes undergo Robertsonian translocation?

Select one:

- submetacentric
- Metacentric
- telocentric
- Holocentric
- acrocentric

Robertsonian translocation is a non-reciprocal (i.e. unequal exchange) that results in a single fused chromosome from 2 acrocentric (non-homologous) chromosomes. A metacentric chromosome has centromere right in the middle. So p and q arms are of equal length. Most chromosomes are submetacentric where the arms are of unequal length. If the centromere is at the tail of a chromosome, it is called telocentric. With holocentric chromosomes, the entire length of the chromosome acts as the centromere. These latter two types are not seen in humans.

The correct answer is: acrocentric

Question 94

Not answered

Marked out of 1.00

Which of the following statements is true about sex linked recessive disorders?

Select one:

- 25% of sons of heterozygous mothers are affected
- 25% of daughters of heterozygous mothers are affected
- Females more affected than males
- All daughters of affected fathers are carriers of the disease
- Affected fathers pass disease to sons

In sex-linked recessive disorders, an affected male can transmit the disease-causing mutation to his daughters, who remains normal phenotypically but carries and transmits the disease-causing allele to her sons. As females have two copies of the X chromosome, they need a double identical mutation for disease expression that is extremely rare.

The correct answer is: All daughters of affected fathers are carriers of the disease

Question 95

Not answered

Marked out of 1.00

The gene implicated in early-onset Alzheimer disease in people with Down syndrome is

Select one:

- Amyloid precursor protein
- Presenilin 1
- Presenilin 2
- Apolipoprotein e3
- Apolipoprotein e4

Explanation:

Individuals with Down syndrome (trisomy 21) have an increased risk of Alzheimer disease (AD) due to genetic factors (amyloid precursor protein gene on chromosome 21 is implicated in early-onset AD). Post-mortem studies (of those over the age of 40) show a high incidence of senile plaques and neurofibrillary tangles as seen in AD. Almost all Down patients above age 40 show neuropathological evidence for AD even if there is no clinical dementia.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 101-102.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 807.

The correct answer is: Amyloid precursor protein

Question 96

Not answered

Marked out of 1.00

Which of the following structures is involved in addictive behaviour?

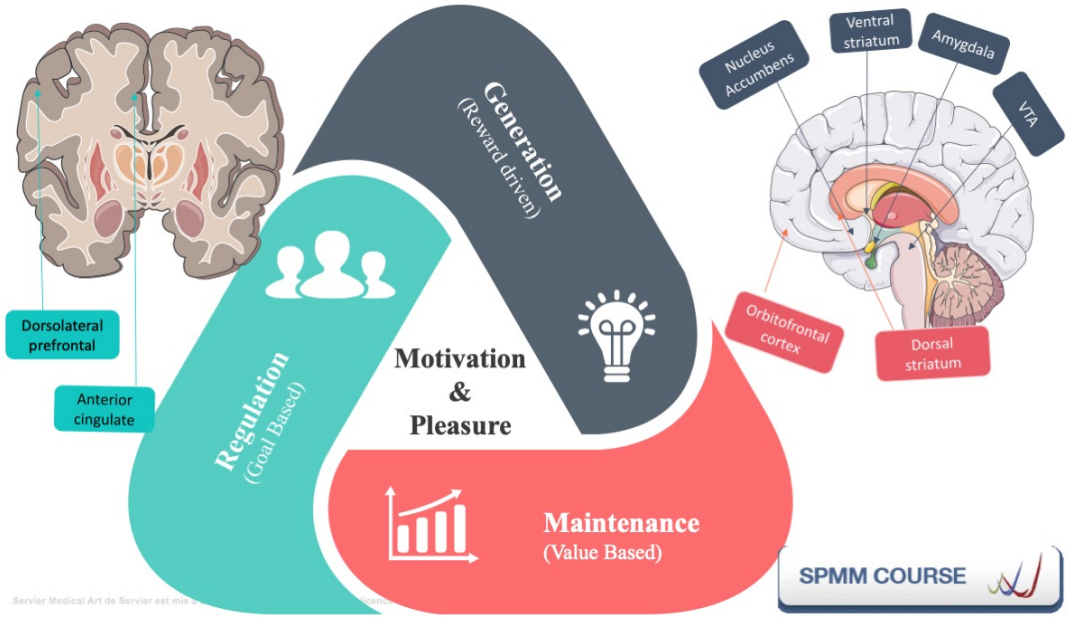
Select one:

- Suprachiasmatic nucleus
- Nucleus tractus solitarius
- Substantia nigra
- Ventral tegmental area
- Preoptic nucleus

Explanation:

The midbrain ventral tegmental area (VTA) lies medial to the substantia nigra and contains dopaminergic neurons that give rise to the mesocorticolimbic dopamine system. These neurons send ascending projections that innervate limbic structures, such as the nucleus accumbens and amygdala; the mesoaccumbens pathways is a central element in the neural representation of reward, and intense research has been devoted to this area. All known drugs of abuse activate the mesoaccumbens dopamine pathway, and plastic changes in this pathway are thought to underlie drug addiction.

For an extended circuitry relevant to addiction see the illustration below:



Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 955.

The correct answer is: Ventral tegmental area

Question 97

Not answered

Marked out of 1.00

Which brain structure is spared during normal aging?

Select one:

- Cerebellum
- Motor neuron
- Cerebral cortex
- Sub thalamic nuclei
- Cranial nerves nuclei

Cranial nerve nuclei are spared during the normal ageing process. All the other structures could be affected.

The correct answer is: Cranial nerves nuclei

Question 98

Not answered

Marked out of 1.00

Which of the following specific phobias has a strongly genetic aetiology?

Select one:

- Spider phobia
- Acrophobia
- Animal phobia
- Space phobia
- Blood injury injection phobia

Blood injury injection phobia is oddly different from other phobias in that the response to exposure is not tachycardia and sympathetically driven heart rate, etc. Instead, a fainting response occurs where the patient may drop fainting with low BP and bradycardia.

The correct answer is: Blood injury injection phobia

Question 99

Not answered

Marked out of 1.00

Loss of taste sensation in the anterior part of tongue is related to

Select one:

- 5th Nerve Palsy
- 9th Nerve Palsy
- 8th Nerve Palsy
- 6th Nerve Palsy
- 7th Nerve Palsy

The facial nerve supplies the taste sensation in the anterior tongue. The chorda tympani (the branch mediating this sensation) leaves the parent nerve, crossing through the middle ear, where it can also be damaged by severe infections, etc.

The correct answer is: 7th Nerve Palsy

Question 100

Not answered

Marked out of 1.00

In which of the following types of inheritance, 75% of children inherit a disease when both parents have the disease and share a similar genotype?

Select one:

- Mitochondrial
- Autosomal dominant
- X linked recessive
- X linked dominant
- Autosomal recessive

If both parents are diseased and have an autosomal recessive condition, then all children will be diseased (100%). If both are diseased with heterozygous genotype, an autosomal dominant disease will be transmitted 3 out of 4 occasions to their children.

The correct answer is: Autosomal dominant

Question 101

Not answered

Marked out of 1.00

Hirano bodies seen in which one of the following types of dementia?

Select one:

- Vascular
- Lewy body type
- HIV-associated
- Frontotemporal
- Alzheimer

Explanation:

Hirano bodies are brightly eosinophilic rod-shaped or elliptical cytoplasmic inclusions that appear to overlap the cell boarder of a neuron cell body. They are mostly found in the hippocampus and are particularly numerous in Alzheimer disease and Pick disease and in patients with the Guam parkinsonism-dementia complex.

Ref: Gray F, Duyckaerts C, de Girolami U. Escourolle and Poirier's Manual of Basic Neuropathology, 6th Edition. 2019. p. 7.

The correct answer is: Alzheimer

Question 102

Not answered

Marked out of 1.00

Tom is a 12-year-old boy with reduced social interactions, poor communication abilities and repetitive stereotypical behaviour. The macroscopic changes involved in the pathogenesis of this condition would be hypoplasia of the

Select one:

- Hippocampus
- Fusiform gyrus
- Cerebellar vermis
- Amygdala
- Angular gyrus

Explanation:

The cerebellum is involved in a broad range of cognitive functions including emotion regulation and social cognition. Abnormalities in the cerebellum are among the most reproducibly reported alterations in autism spectrum disorder. Differences in the volume, as compared to healthy people, of Crus I, cerebellar vermis, and anterior cerebellum have been reported in individuals with autism spectrum disorder.

Ref: Laidi C et al. Cerebellar anatomical alterations and attention to eyes in autism. Scientific reports. 2017 Sep 20;7(1):1-11.

The correct answer is: Cerebellar vermis

Question 103

Not answered

Marked out of 3.00

Dangerous complications of eating disorders

Identify the metabolic abnormalities commonly seen in the following clinical situations

A 16-year-old girl with history of significant weight loss, body image distortions and morbid fear of fatness had shown some endocrine abnormalities in her blood

A 23-year-old woman with history of bulimia nervosa and is determined to reduce weight by repetitive vomiting

A 20-year-old woman uses regular laxatives for last two years in order to reduce weight. In addition to hypokalemia associated with acidosis, what other electrolyte abnormalities results from chronic laxative abuse?

Hyperhomocustinemia
Hypophosphatemia
Low T3, high growth hormone and cortisol
Low cortisol, high T3 and growth hormone
Hyponatremia
Hypothyriodism
Hyperkalemia
Hypercortisolemia
Hypokalemia
Hyperphosphatemia.

Explanation: The diagnosis is anorexia nervosa. The common endocrine changes in anorexia include low concentrations of luteinising hormone, follicle stimulating hormone and oestradiol, low T3 but T4 in low normal range, average concentrations of thyroid stimulating hormone (low T3 syndrome), increase in plasma cortisol and raised growth hormone level.

Electrolyte disturbances can be of variable nature and are present in those who frequently vomit or misuse large quantities of laxatives or diuretics. Vomiting results in metabolic alkalosis and hypokalemia. In repetitive vomiting, loss of hydrochloric acid from gastric juices lead to metabolic alkalosis (loss of acid - alkalosis). Laxative misuse results in metabolic acidosis, hyponatraemia, hypokalemia. During laxative induced diarrhoea, a considerable amount of bicarbonate may be lost in the stool. With healthy kidneys, the lost bicarbonate is replaced adequately, and a serious base deficit does not develop. When there is poor renal blood flow due to hypovolaemia/starvation, base deficit and acidosis develop rapidly. Acidosis also results from excessive production of lactic acid when patients have severe diarrhoea.

Ref: Støving RK. Mechanisms in endocrinology: Anorexia nervosa and endocrinology: A clinical update. European journal of endocrinology. 2019 Jan 1;180(1):R9-27.

The correct answer is: A 16-year-old girl with history of significant weight loss, body image distortions and morbid fear of fatness had shown some endocrine abnormalities in her blood → Low T3, high growth hormone and cortisol, A 23-year-old woman with history of bulimia nervosa and is determined to reduce weight by repetitive vomiting → Hypokalemia, A 20-year-old woman uses regular laxatives for last two

years in order to reduce weight. In addition to hypokalemia associated with acidosis, what other electrolyte abnormalities results from chronic laxative abuse? → Hyponatremia

Question 104

Not answered

Marked out of 3.00

Features of defence mechanisms

For each of the characteristic features described below chose the most appropriate defence mechanism suitable from the abov

Ascribing unacceptable motivations to others	Choose...
Resentment by noncooperation	Choose...
Failure to accept or face conflicts	Choose...

Projection
Passive aggression
Turning into self
Identification
Humour
Denial
Altruism
Acting out
Repression
Suppression
Regression

Explanation:

Projection is a primitive or immature defense mechanism that involves attributing one's own internal unacceptable ideas and impulses to ar external target, such as another individual, and reacting accordingly to them. It includes severe prejudice, rejection of intimacy through suspiciousness, hypervigilance to external danger, and injustice collecting.

Passive aggression is aggression toward an object expressed indirectly and ineffectively through passivity, masochism, and turning against the self.

Seeing, but refusing to acknowledge what one sees, or hearing and negating what is heard are examples of denial. Denial avoids becoming aware of some painful aspect of reality.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 892.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 1066.

The correct answer is: Ascribing unacceptable motivations to others → Projection, Resentment by noncooperation → Passive aggression, Failure to accept or face conflicts → Denial

Question 105

Not answered

Marked out of 3.00

Supportive techniques

For each of the following examples, identify the type of supportive techniques aimed at eliciting information

Sometimes when people are upset and low in their mood, they think of hurting themselves. Has this been true for you?

I think it is best for you to consider psychotherapy at this time. If I am you, I will give this a serious thought

This depression may be very difficult for you. But I think it is very likely with the proper treatment you can get back to your job.

Validation
Disapproval
Acknowledgment of affect
Statement of respect
Positive reinforcement
Advice
Postponement
Reassurance
Partnering

Explanation:

Validation/normalization: It helps to decrease a patient's sense of embarrassment about a feeling or behaviour. It can be done by quoting how it is normal for people to have different emotions/ reactions/ behaviours, etc.

Advice: Many patients seek help directly; it is acceptable to provide advice but based on sound understanding of the context. Premature advice can be obstructive than facilitative.

Ref: Adapted from Kay J & Tasman A. Essentials of Psychiatry, 2nd edition, 2006. John Wiley & Sons, Ltd.

The correct answer is: Sometimes when people are upset and low in their mood, they think of hurting themselves. Has this been true for you? → Validation, I think it is best for you to consider psychotherapy at this time. If I am you, I will give this a serious thought → Advice, This depression may be very difficult for you. But I think it is very likely with the proper treatment you can get back to your job. → Reassurance

Question 106

Not answered

Marked out of 4.00

Rating scales and their uses

For each description given below, select the relevant rating scale from the list above;

This scale is useful for rating severity in patients with a pre-existing diagnosis of OCD — Cho

Commonly used screening tool used in primary care and general population studies — Cho

This scale measures overall psychosocial functioning — Cho

7-point scale useful to assess global observation of severity of psychiatric illness — Cho

- Yale-Brown obsessive compulsive scales (YBOCS)
- Hospital anxiety and depression scale (HADS)
- Young mania rating scale (YMRS)
- Clinical Global impressions scale (CGI)
- Positive and negative symptoms scale (PANSS)
- Global assessment of functioning scale
- Brief psychiatric rating scale (BPRS)
- General health questionnaire (GHQ)
- Mini-mental state examination (MMSE)
- Minnesota multiphasic personality inventory (MMPI)

Explanation:

The Yale-Brown Obsessive-Compulsive Scale (Y-BOCS) is a clinician-administered semi-structured interview allowing the rating of severity in patients with a pre-existing diagnosis of OCD.

The General Health Questionnaire (GHQ) is a self-rated questionnaire used as a screening instrument for the presence of psychiatric illness. The patient is asked to report the presence of a list of symptoms in the preceding weeks. Four versions are available, using 12, 28 (max score = 84), 30, and 60 items.

The Global Assessment of Functioning Scale (GAF) is a rating scale measuring overall psychosocial functioning. The maximum score on the GAF is 100.

The Clinical Global Impression (CGI) scales were developed as simplified measures to reflect the clinician's overall impression of a patient's condition (CGI-Severity rated 1-7 from 'normal' to 'among the most extremely ill') and change over time (CGI-Improvement, CGI-I, rated from 'very much improved' to 'very much worse').

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. pp. 96-97.

Dunlop BW et al. Transdiagnostic clinical global impression scoring for routine clinical settings. Behavioral Sciences. 2017 Sep;7(3):40.

The correct answer is: This scale is useful for rating severity in patients with a pre-existing diagnosis of OCD → Yale-Brown obsessive compulsive scales (YBOCS), Commonly used screening tool used in primary care and general population studies → General health questionnaire (GHQ), This scale measures overall psychosocial functioning → Global assessment of functioning scale (GAF), 7-point scale useful to assess global observation of severity of psychiatric illness → Clinical Global impressions scale (CGI)

Question 107

Not answered

Marked out of 3.00

Adverse effects

For each of the following drugs, identify the most important side effect associated with its therapeutic use

Procyclidine	Choose...
Ziprasidone	Choose...
Olanzapine	Choose...

Diastolic Hypertension
Nasal Stiffness
Glucose dysregulation
Prolactin elevation
Amenorrhoea
Blurred vision
Bradycardia
Polydipsia
Weight Loss
Hypersalivation
QTc prolongation

Procyclidine, Benztropine, Orphenadrine and Biperiden are anticholinergic drugs that can cause dry mouth, constipation, urinary retention and blurred vision. The latter is associated with impaired visual accommodation, causing difficulty in reading.

There are concerns over the use of Ziprasidone, as it can increase QTc. However with this drug, EPSE is equivalent to placebo, no appreciable weight gain, and no appreciable anticholinergic side effects are noted.

Although possible glucose dysregulation and metabolic syndrome are reported with most atypical antipsychotics, Olanzapine is the most common offender. Metabolic syndrome is a cluster of disorders comprising of obesity(central and abdominal), dyslipidaemias, glucose intolerance, insulin resistance (or hyperinsulinaemia) and hypertension.

The correct answer is: Procyclidine → Blurred vision, Ziprasidone → QTc prolongation, Olanzapine → Glucose dysregulation

Question 108

Not answered

Marked out of 3.00

Research studies in genetics

Select the most appropriate type of study for each description given below

An investigator is interested in calculating the risk of illness within families among first, second and third degree relatives and compare the rates with the established risk in the general population.

An investigator wants to compare the likelihood of the observed rate of illness in a given pedigree with those families that have a differing mode of transmission

A researcher has no candidate genes of a priori interest but wants to study the whole genome in large samples of patients and controls to investigate the most commonly observed polymorphic variations in the patients.

Genome-wide association studies
Outcome studies
Pedigree studies
Transgenic studies
Adoption studies
Treatment studies
Population studies
Segregation analysis
Family studies
Association studies

Family studies aid in determining the likelihood of illness among relatives. **Segregation analysis** is used to investigate whether familial data for a disorder or trait suggests any particular modes of inheritance. Both monogenic (Mendelian) or polygenic models can be tested in this fashion. In addition, age of onset and sex effects can be taken into account besides the specific genetic hypothesis under consideration. In **Genome-Wide Association Studies** (GWAS), a 'candidate gene' approach is not used. Instead, several thousands of single nucleotide polymorphisms are assayed in thousands of individuals.

The correct answer is: An investigator is interested in calculating the risk of illness within families among first, second and third degree relatives and compare the rates with the established risk in the general population. → Family studies, An investigator wants to compare the likelihood of the observed rate of illness in a given pedigree with those families that have a differing mode of transmission → Segregation analysis, A researcher has no candidate genes of a priori interest but wants to study the whole genome in large samples of patients and controls to investigate the most commonly observed polymorphic variations in the patients. → Genome-wide association studies

Question 109

Not answered

Marked out of 4.00

Clinically notable motor disorders

Choose one option for each of the following descriptions

Sustained abnormalities of posture occurring focally in writer's cramp or torticollis

Brief jerky downward movements of the wrist, when the patient extend both arms with the wrists dorsiflexed, palms facing forward, and eyes closed

Quasi-purposeful movements affecting multiple joints with a distal preponderance

Quivering of the muscle seen under the skin in neuromuscular disorder

Chorea
Tardive Dyskinesia
Myoclonus
Rigidity
Hemibalismus
Fibrillation
Fasiculations
Dystonia
Tics
Asterixis

Explanation:

Writer's cramp is a form of focal dystonia. Dystonia is an extrapyramidal motor disturbance consisting of slow, sustained contractions of the axial or appendicular musculature. These include torticollis, trismus, blepharospasm, oculogyric crisis, and opisthotonus. In contrast, tics are spontaneous, repetitive, rhythmic movements that are semi-voluntary (may be resisted for a short time with difficulty).

Asterixis can be elicited by having the patient extend both arms with the wrists dorsiflexed and palms facing forward and eyes closed. Brief jerky downward movements of the wrist are considered a positive sign. Asterixis is commonly seen with metabolic encephalopathies.

The term chorea means dance. Chorea is the sudden and involuntary movement of several muscle groups, with the resultant action appearing like part of a voluntary movement.

Fasciculations may be seen under the skin as quivering of the muscle. Although fasciculations are typically benign (particularly when they occur in the calf), if widespread, they can be associated with neuromuscular disease, including amyotrophic lateral sclerosis (ALS).

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 1138.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. pp.104, 1016.

Neurological History and Physical Examination - Medscape, http://emedicine.medscape.com/article/1147993-overview (accessed May 5, 2021)

The correct answer is: Sustained abnormalities of posture occurring focally in writer's cramp or torticollis → Dystonia, Brief jerky downward movements of the wrist, when the patient extend both arms with the wrists dorsiflexed, palms facing forward, and eyes closed → Asterixis, Quasi-purposeful movements affecting multiple joints with a distal preponderance → Chorea, Quivering of the muscle seen under the skin in neuromuscular disorder → Fasciculations

Question 110

Not answered

Marked out of 3.00

Bedside tests of cognition

For each of the following options below, choose the most appropriate test from the given list:

A test of visual memory where a child is shown ten geometric designs, one at a time, and asked to reproduce each one as exactly as possible on plain paper from memory.

A test to check inhibitory control aspect of executive function.

This test measures the recall of learned material. It is a complex figure test, which distinguishes between brain damage and mental retardation.

Raven's progressive matrices
Benton test
Stanford-Binet test
Stroop test
National adult reading test
Vocalbulary test
Trail making Test
Wisconsin card sorting test
Rey-Osterreith test
Digit span test

Explanation:

The Benton Visual Retention Test assesses memory for 10 geometric designs after 10-second exposure. It is often found useful in detecting intellectual disability.

The Stroop effect is the delay and disruption in naming the colours of words printed in non-matching coloured ink, as when the word 'red' is printed in blue ink, the word 'blue' in green ink, and so on. To perform the task it is necessary to ignore the meanings of the printed words and to respond only to the colours in which they are printed, but with experienced adult readers automatic processing tends to occur, causing interference in the processing of information and a significant increase in the time required to name the colours, compared with the time taken to name the colours of meaningless strings of letters. The Stroop Effect is utilised in testing the executive faculty of inhibition (suppression of interferences) and error correction.

The Rey-Osterrieth Complex Figure test examines the ability to draw and later recall a complex geometric configuration. It tests visual memory as well as executive deficits in the development of strategies and planning. Performance is impaired in patients with dementia and amnesic syndrome.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 53.

Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015

The correct answer is: A test of visual memory where a child is shown ten geometric designs, one at a time, and asked to reproduce each one as exactly as possible on plain paper from memory. → Benton test, A test to check inhibitory control aspect of executive function. → Stroop test, This test measures the recall of learned material. It is a complex figure test, which distinguishes between brain damage and mental retardation. → Rey-Osterreith test

Question 111

Not answered

Marked out of 5.00

Psychodynamic reactions

Identify the defence mechanisms involved in each of the following situations:

A 39-year-old woman is diagnosed with metastatic cervical cancer. She exhibits no emotions with respect to her physical health but discusses at great length about the pathology of cervical cancer and anatomy of cervix.

A 34-year-old man is served debt collection notice for unpaid bills. He does not react to this note but soon gets furious at his wife for no apparent fault of hers.

A 20-year-old woman is diagnosed with borderline personality disorder. She finds it difficult to trust people and cannot accept that people can have both good and bad qualities. She thinks that everyone in this world must be either good or bad with no moderation in between.

The above-described woman undergoes psychotherapy. She calls her therapist as "the best person in the world" within the first few weeks of uninterrupted sessions.

The same woman calls her therapist the "the worst person in the world" and demands a change in therapist when the therapist announces a short break in sessions due to a planned holiday.

Omnipotence
Reaction
Formation
Intellectualisation
Repression
Idealisation
Sublimation
Displacement
Denial
Denigration
Regression
Suppression
Rationalisation
Isolation
Splitting

Explanation:

Intellectualisation is characterised by focusing on abstract concepts, logic, and other forms of intellectual reasoning to avoid facing painful emotions (e.g. a victim of a traumatic abuse experience may discuss the statistics of abuse instead of talking about the particular experience).

Displacement is the process by which interest and/or emotion is shifted from one object onto another less threatening one. For example, the person who is having problems at work may displace the anger felt for their boss onto their family by displaying irritability and moodiness at home.

Functionally, patients with borderline personality disorder distort their relationships by considering each person to be either all good or all bad. They see people as either nurturing attachment figures or as hateful, sadistic figures who deprive them of security needs and threaten them with abandonment whenever they feel dependent. As a result of this **splitting**, the good person is **idealised,** and the bad person

devalued and **denigrated**.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p.893.

Casey P, Kelly B. Fish's clinical pathology: Signs and symptoms in psychiatry, 4th Edition. 2019.p. 129.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 564.

The correct answer is: A 39-year-old woman is diagnosed with metastatic cervical cancer. She exhibits no emotions with respect to her physical health but discusses at great length about the pathology of cervical cancer and anatomy of cervix. → Intellectualisation, A 34-year-old man is served debt collection notice for unpaid bills. He does not react to this note but soon gets furious at his wife for no apparent fault of hers. → Displacement, A 20-year-old woman is diagnosed with borderline personality disorder. She finds it difficult to trust people and cannot accept that people can have both good and bad qualities. She thinks that everyone in this world must be either good or bad with no moderation in between. → Splitting, The above-described woman undergoes psychotherapy. She calls her therapist as "the best person in the world" within the first few weeks of uninterrupted sessions. → Idealisation, The same woman calls her therapist the "the worst person in the world" and demands a change in therapist when the therapist announces a short break in sessions due to a planned holiday. → Denigration

Question 112

Not answered

Marked out of 3.00

Mechanism of drug action

Find the drug that matches each of the following mechanisms of action;

Phosphodiesterase-5 inhibitor	Choose...
Butyrylcholinesterase inhibitor	Choose...
Dopamine reuptake inhibitor	Choose...

Donepezil
Agomelatine
Aripiprazole
Galatamine
Bupropion
Duloxetine
Sildenafil
Mirtazapine
Rivastigmine
Reboxetine

Explanation:

Sildenafil is a phosphodiesterase-5 inhibitor.

Rivastigmine is a cholinesterase inhibitor. It is both an acetylcholinesterase inhibitor and a butyrylcholinesterase inhibitor.

Bupropion is a noradrenergic and dopaminergic reuptake inhibitor. It is indicated for depression but is only licensed in the UK for treatment of nicotine dependence.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 699, 729.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 290.

The correct answer is: Phosphodiesterase-5 inhibitor → Sildenafil, Butyrylcholinesterase inhibitor → Rivastigmine, Dopamine reuptake inhibitor → Bupropion

Question 113

Not answered

Marked out of 4.00

EEG Changes

For which of the following psychiatric conditions, identify the EEG changes most l

Angelman's syndrome	Choose...
Infantile spasms	Choose...
Neurosyphilis	Choose...
Stroke	Choose...

Focal or regular delta activity.
Episodic discharges recurring every 1-3 seconds.
Generalised multiple spike-wave discharges.
High amplitude 2-3Hz frontal activity wiht superimposed epileptiform discharges.
Diffuse giant waves with a chaotic background of multifocal spikes and sharp waves.
Periodic generalised 1-2Hz sharp waves.
Triphasic waves.
Non-specific increase in slow waves.
Flattened trace

In Angelman syndrome, EEG changes are notable by the age of 2. There are prolonged runs of high amplitude 2-3 Hz frontal activity with superimposed inter-ictal epileptiform discharges in patients of all ages. There is occipital high amplitude rhythmic 4-6 Hz activity facilitated by eye closure, which is seen under the age of 12. There is no difference in EEG patterns between AS patients with or without seizures.

In infantile spasms (seen in tuberous sclerosis), the EEG finding is described as hypsarrhythmia. Diffuse high voltage waves (>400 microvolts) with irregular, chaotic multifocal spikes and sharp waves in the background is the characteristic feature of hypsarrythmia. Clinical seizures are associated with a marked suppression of the background; this is called the electrodecremental response.

In neurosyphilis, there is a non-specific increase in slow waves occurring diffusely over the scalp.

In stroke, there is focal or regional delta activity.

The correct answer is: Angelman's syndrome → High amplitude 2-3 Hz frontal activity with superimposed epileptiform discharges, Infantile spasms → Diffuse giant waves with a chaotic background of multifocal spikes and sharp waves, Neurosyphilis → Non-specific increase in slow waves, Stroke → Focal or regional delta activity

Question 114

Not answered

Marked out of 4.00

MoK Pharmacology CXEMI004

Pharmacokinetics of antipsychotics

Identify the antipsychotic drug using descriptions below;

1. Half life more than 50 hours ✖

2. Antipsychotics with a fast dissociation profile (choose two drugs) ✖

3. D2 and D3 antagonist, with weak 5HT-7 antagonist actions

Chlorpromazine
Promazine
Haloperidol
Sertindole
Olanzapine
Risperidone
Aripiprazole
Amisulpride
Clozapine
Quetiapine

Explanation:

The half-life of antipsychotics varies greatly. Shorter half-life medications, such as quetiapine and ziprasidone, have 7-12 hours of half-life. Risperidone has a 3-hour half-life, but it is metabolized to 9-hydroxy risperidone (aka paliperidone), which has a 24-28-hour half-life while olanzapine has a 30-hour half-life. The longest half-life, about three days, is seen with more recently introduced agents aripiprazole, brexpiprazole and cariprazine. The mean elimination half-life is 75 hours for aripiprazole and 94 hours for the major metabolite dehydro-aripiprazole.

See below for a table.

Clozapine has relatively low potency as a D2 receptor antagonist. At therapeutic doses, clozapine occupies fewer D2 receptors than the other drugs that treat psychosis. Clinically effective doses of clozapine occupy only 40-50% of striatal D2 receptors. Similarly, the receptor antagonism for quetiapine is generally lower than that for other antipsychotic drugs. This is probably why clozapine and quetiapine have low propensity to cause EPS. It has also been postulated that clozapine binds more loosely to the D2 receptor, and because of this ‘fast dissociation’, more normal dopamine neurotransmission is possible. Quetiapine also shares this unique fast dissociation profile with clozapine, leading to the description of ‘hit-and’run mechanism’ ('hit' the D2 receptor (blockade) only for a short while to cause antipsychotic effects and then 'run' (dissociate) before they cause extrapyramidal side effects). As a result, they are recommended for use in Parkinson's disease with psychosis.

In addition to its actions at D2 receptors, amisulpride has some D3 antagonist actions and some weak 5HT-7 antagonist actions, which may explain some of its negative symptom and antidepressant actions.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 57.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 613.

Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 223.

The correct answer is:
Half life more than 50 hours - Aripiprazole

Antipsychotics with a fast dissociation profile (choose two drugs) - Clozapine, Quetiapine

D2 antagonistic effect - Amisulpride

Half-life values for oral atypical antipsychotics

Drug	Estimated $t_{1/2}$
Cariprazine	• 72-96 hours; active metabolites – up to 3 weeks
Brexpiprazole	• 91 hours
Aripiprazole	• 72 hours
Olanzapine	• 30 hours
Paliperidone	• 24-28 hours
Clozapine	• 16 hours
Amisulpride	• 12 hours
Quetiapine/Ziprasidone	• 7 hours
Risperidone	• 3 hours

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Question 115

Not answered

Marked out of 6.00

MoK Psychology CXEMI 005

Cloninger's Theory of Personality

Identify the correct behavioural traits to match the dimensions of temperament described by Cloninger.

1. Harm avoidance traits (Choose TWO) ✖ ✖
2. Novelty seeking traits (Choose TWO) ✖ ✖
3. Reward dependence traits (Choose TWO) ✖ ✖

Explanation: Cloninger's psychobiological model of personality includes four dimensions of temperament (each 50 to 60 % heritable), which manifest early in life and 3 components of character, which are shaped by environment. The temperamental dimensions include

- (1) Novelty-seeking (includes frustration avoidance, impulsive decision-making)
- (2) Harm-avoidance (pessimistic worry about the future, passive avoidant behaviour, fear of uncertainty);
- (3) Reward-dependence (sentimentality, social attachment, and dependence on praise and approval)
- (4) Persistence (high perseverance and tolerance of frustration). The character dimensions are self-directedness, cooperativeness, and self-transcendence.

The correct answer is:

Harm avoidance traits (Choose TWO)- Fear of uncertainty, Pessimism,

Novelty seeking traits (Choose TWO)- Frustration Avoidance, Impulsivity,

Reward dependence traits (Choose TWO)- Praise dependence, Sentimentality

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